Where the Windfall Goes: Fossil Revenue, Domestic Absorption and Carbon Emissions

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Abstract

Increases in the world price of oil, gas, or coal generate a windfall for producer economies, and the energy transition is likely to make these windfalls more volatile. Using national accounts, government revenues and sector-level fossil CO_2 for 163 countries between 1981 and 2024, I exploit the interaction between predetermined fossil rents and unexpected changes in world fuel prices to construct a windfall shock. In the economies with resource revenue data, windfalls accrue almost entirely to the government budget. Windfalls are mostly saved abroad in the first year but spent on consumption, investment and imports within two years. Each dollar of windfall revenue increases domestic CO_2 by 0.52 kg (95 per cent weak-instrument-robust interval: 0.20 to 0.89), around half the carbon intensity of an average dollar of gross domestic product, mostly through road transport, power and buildings. Cement emissions rise through higher utilisation of existing kilns, without new capacity. Instrumenting the windfall using oil supply news from announcements by the Organization of the Petroleum Exporting Countries, I find that the response of macroeconomic variables, construction, buildings and road transport CO_2 is of the same magnitude as in the baseline. In independent energy statistics, the instrumented increase in energy CO_2, averaged over years zero to five, is equally large. Positive windfall shocks increase emissions in cement, buildings and power, but negative shocks of equal size do not decrease them. Over the 2003 to 2014 commodity boom, windfalls added around 2.7 billion tonnes of CO_2, with climate damage of roughly $500 billion.

Keywords: fossil fuel windfalls; oil supply news; carbon emissions; transition risk; domestic absorption; fiscal policy
JEL codes: Q32, Q54, E62, F41, H54, O13

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1 Introduction

Fluctuations in world prices of fossil fuels translate into substantial changes in the revenues of energy-producing countries, through taxes, royalties and national oil companies. A recent literature has shown that higher prices reduce carbon emissions in consuming countries: carbon taxes in European countries have lowered emissions with no measurable effect on output (Metcalf and Stock 2023), and carbon price shocks reduce both emissions and economic activity (Känzig 2023). Little is known, however, about the effect of these changes in revenues on carbon emissions in producing countries, through the local spending of revenues on energy, vehicles, construction and public services. This paper investigates the carbon footprint of that spending.

This question is particularly salient in light of the ongoing energy transition, which is expected to reduce fossil fuel rents in these countries as demand falls, except for temporary spikes during supply disruptions, as in 2022, when the world price of crude oil increased by about 40 per cent and that of coal about two and a half times. The growing literature on transition risk (e.g. Battiston et al. 2017; Mercure et al. 2018) includes studies of the effect of the expected decrease in revenues on asset prices, sovereign debt, and economic growth in these countries (e.g. van der Ploeg and Rezai 2020; Peszko et al. 2020). A recent study shows that commodity price booms precede faster growth in carbon emissions per capita across countries (Oyekola et al. 2024). However, it is not clear whether the increases in emissions are due to higher revenues, or due to the higher global demand that is likely to increase both carbon emissions and prices. Nor is it known which sectors the extra emissions come from, how large the marginal effect of revenues on carbon emissions is, or whether the increase is reversed when energy prices fall. This paper investigates these questions.

I exploit variation in windfall changes in fossil fuel rents, constructed as the product of each country's past share of oil, gas and coal rents in GDP (averaged over 6 to 10 years before the shock) and the surprise component in the evolution of real world fuel prices (estimated using only past prices of each fuel). Including year effects makes this approach a shift-share regression, comparing countries with different shares of fuel rents in GDP in years with different surprises in world fuel prices (Goldsmith-Pinkham et al. 2020; Borusyak et al. 2022). Since world fuel prices may be correlated with global demand (Kilian 2009; Baumeister and Hamilton 2019), I also instrument windfall changes in rents with the oil supply news shocks constructed by Känzig (2021), who exploits variation in oil futures prices around announcements by the Organization of the Petroleum Exporting Countries (OPEC) to identify news about oil supply. I study the effect of windfall changes in rents on national accounts and government revenues, and on fossil carbon dioxide (CO_2) emissions

across all sectors of the Emissions Database for Global Atmospheric Research (EDGAR), using the local projections method of Jordà (2005), for a panel of 163 countries (the 147 outside OPEC in the instrumented estimates). I then check the results on carbon emissions with independent energy statistics, and construct estimates of the carbon content of windfall revenues by instrumenting the cumulative change in rent revenues with the windfall shock.

Most windfall revenue ends up in the government budget, and it is saved first and spent later. A one-unit increase in the windfall shock increases fossil rent by 0.81 of GDP in the year of the shock, close to the benchmark of 1 implied by the construction of the shock. The increase in rent translates into an increase in government revenues from resource rent. In the countries with data on resource rent revenues, government revenues from resource rent rise by an estimated 0.76 of the increase in rent over three years and by about the full increase over five years. The increase in revenues is reflected in a large increase in the current account balance of 0.64 of GDP in the year of the shock, implying that most of the revenues are saved abroad, although the increase in the current account balance disappears after two years. Over the following years, the shock raises household and government consumption, public and total fixed investment, and imports, which are still higher eight years after the shock, significantly so for all but public investment. For a one-standard-deviation windfall in rent economies (economies in which rent is above one-tenth of GDP), real GDP increases by 3.2 per cent after five years, household consumption by 3.8 per cent, government consumption by 7.4 per cent, and fixed investment by 12.5 per cent. The effects on the current account balance and on real GDP, household consumption, and public investment are of similar magnitude when the windfall is instrumented with supply news shocks.

Windfalls raise CO_2 emissions: as windfalls are absorbed, total fossil fuel CO_2 rises by 0.27 log points per unit shock over a five-year horizon. For a typical windfall, that is 2.3 per cent of emissions, and emissions from road transport and buildings rise by around 6.5 per cent and 8 per cent. Emissions from extraction do not respond, as one would expect for price windfalls when production volumes are set by capacity. A dollar of windfall revenue raises domestic CO_2 by 0.52 kg over nine years, and by between 0.20 and 0.89 kg in the 95 per cent Anderson-Rubin confidence set, which is robust to a weak instrument. This is roughly 50 per cent of the carbon intensity of the average dollar of GDP in economies with rents above 5 per cent of GDP, since much of the revenue is saved or spent on imports. Road transport, power, and buildings make up about 60 per cent of the carbon content of a dollar of windfall revenue, and construction materials, mostly cement, make up another 7 per cent.

The response of total emissions is robust to including global cycle controls and region-by-year effects (0.27 and 0.24 after five years, significant at the 10 per cent level or better), lies in the interval [0.21, 0.31] and remains significant at the 5 per cent level when excluding

any one year of price surprises, is robust to excluding the whole boom of 2003 to 2014, and has a p-value of 0.018 when standard errors are calculated at the level of the 105 fuel-year price surprises that drive the identifying variation. Results hold in the US Energy Information Administration's (EIA) independently compiled energy statistics, which are based on measured fuel usage: the response is larger, and the carbon content of a dollar of revenue is about the same (0.51 against 0.52 kg). In the EIA data, windfalls instrumented with supply news raise energy CO_2 emissions by as much as price surprises do on average over years 0 to 5. In EDGAR, windfalls instrumented with supply news raise total CO_2 emissions by at least as much as price surprises in the first two years, but the response fades thereafter. Oil and gas windfalls have similar carbon intensities (0.50 and 0.74 kg per dollar; equality is not rejected at the 5 per cent level), despite their prices reaching domestic fuel markets through different channels.

Cement emissions make up a small part of the carbon footprint of windfall revenues but give a cleaner measure of the investment channel than any aggregate. Cement is produced locally, its process emissions per tonne are set by chemistry, it is available in two independent emissions datasets, and it is measured in tonnes. Cement emissions respond strongly, rising by 0.91 log points per unit shock over a five-year horizon, faster than GDP. Cement production also rises by at least as much, but countries do not commission additional clinker production capacity after windfalls. The investment response operates through the utilisation of existing kilns, as one might expect for heavy industry with putty-clay capital, where capacity and emission rates are locked in once a plant is built.

Windfalls also generate asymmetric responses: positive windfalls raise emissions in cement, buildings and power and raise household consumption, but negative windfalls of equal size do not reduce them. In line with the literature on fiscal policy procyclicality (Kaminsky et al. 2004; Frankel et al. 2013), public investment falls after negative windfalls. I call this asymmetry the carbon ratchet, and it mirrors at the sector level the finding that aggregate emissions rise more in expansions than they fall in contractions of equal size (York 2012; Sheldon 2017). Despite the asymmetric response in these sectors, the response of total emissions is not significantly asymmetric. Fiscal rules, autocracy, and fuel underpricing are each associated with a smaller response of public investment two years after the shock, but not with a smaller response of cement. Fuel underpricing, if anything, is linked to a larger total emissions response. Applying the common responses to each country's shocks, I estimate that windfalls during the commodity boom of 2003 to 2014 increased fossil fuel CO_2 emissions by about 2.7 Gt, 3.5 per cent of emissions in the 32 economies in which resource rents exceeded 10 per cent of GDP in at least one year of the boom. Valued at the mean social cost of carbon from Rennert et al. (2022), these emissions imply damage of roughly $500

billion, or 10 cents for each dollar of windfall revenue.

1.1 Related literature

This paper contributes to three bodies of literature. The first is the literature on macroeconomic effects of oil and commodity price shocks. The recent contribution by Känzig (2021) focuses on identifying oil supply news shocks from announcements by OPEC. Before that, the literature mainly focused on separating oil supply shocks from oil demand shocks (Kilian 2009; Baumeister and Hamilton 2019). Other contributions have highlighted that commodity prices are important drivers of business cycles in emerging markets (Fernández et al. 2017; Drechsel and Tenreyro 2018). Arezki et al. (2017) investigate the impact of news shocks by exploiting giant oil discoveries, while Cavalcanti et al. (2015) and Mohaddes and Pesaran (2017) analyse how swings in commodity and oil prices affect economic growth. In this paper, I treat supply news as an external instrument, in the terminology of Stock and Watson (2018), alongside price surprises in a shift-share design, in the economies that produce these commodities. I trace the evolution of revenues into the government budget and the national accounts, and examine the effects on emissions by sector.

The paper also contributes to a literature that examines how commodity booms affect economic growth and carbon emissions. The closest study is by Oyekola et al. (2024), who find that booms in commodity prices increase carbon emissions in democratic countries. They relate the growth of CO_2 per capita to the change in an export-weighted index of 40 commodity prices. Using a different shock, sample and estimator, I confirm the aggregate finding of Oyekola et al. (2024), and account for its timing. In particular, Oyekola et al. (2024) find that commodity price increases have no contemporaneous effect on emissions but a positive effect one year after the shock. This pattern is consistent with revenues being saved in the year of the shock and then being spent with a lag. A related contribution by Friedrichs and Inderwildi (2013) shows that economies that are rich in carbonaceous fuels are more carbon intensive. This paper goes one step further and examines the transmission channel. More generally, a large literature has examined the procyclicality of carbon emissions, and how they depend on the source of economic growth (e.g. Doda 2014; Burke et al. 2015; Khan et al. 2019). A separate literature has estimated the macroeconomic effects of carbon prices and temperature (Metcalf and Stock 2023; Känzig 2023; Bilal and Känzig 2024). The present paper shows that increases in fossil fuel prices raise emissions in the countries that sell those fuels, via the revenue channel, partly offsetting the reductions that the same price increases bring about among buyers.

The paper further links the normative and empirical literature on how resource wind-

falls should be managed with research on carbon lock-in. Theoretical papers recommend saving the windfall abroad and gradually investing it domestically (van der Ploeg and Venables 2011; Collier et al. 2010; van der Ploeg 2011). In contrast, the empirical literature documents procyclical fiscal policy in commodity-exporting countries (Céspedes and Velasco 2014; Frankel et al. 2013). Ilzetzki et al. (2013) document important differences in fiscal multipliers across exchange rate regimes and with the degree of openness and public debt. The spending of windfalls has been associated with democracy and external debt (Arezki and Brückner 2012) and with Dutch disease (Corden and Neary 1982), although in US resource booms manufacturing as a whole is not crowded out (Allcott and Keniston 2018). Another literature has measured the cost of fuel subsidies (Davis 2014; Coady et al. 2017; Ross et al. 2017). Turning to the climate economics literature, many papers have shown that capital is long-lived and therefore locks in future emissions (Unruh 2000; Davis et al. 2010; Seto et al. 2016; Tong et al. 2019). The present paper shows that the spending response has a quantifiable carbon footprint, that the composition of this response varies with the fiscal rules in place and the way fuel prices are set, and that the increase in emissions in cement, buildings and power during a boom is not reversed in a subsequent bust. Thus, there is a link between the transition risk faced by producer economies and the amount of carbon they lock in.

Section 2 presents the framework, and Sections 3 and 4 describe the data and the empirical strategy. Section 5 reports the results: it traces out the transmission and the carbon footprint, presents additional evidence based on supply news, examines the construction channel and the identification, and studies the asymmetric effect of positive and negative windfalls and the role of institutions. Section 6 computes the climate cost of windfalls, and Sections 7 and 8 discuss and conclude.

2 Framework

2.1 Absorption

Let R_t be the fossil rent revenue in period t and let $\widetilde{R}_t = R_t - E_{t-1}R_t$ be the windfall. The economy can save the windfall abroad, consume it, or invest it. Domestic absorption is domestic spending on consumption and investment, on both domestic and imported goods. I denote the share of the windfall saved abroad by σ and the shares of the remaining windfall absorbed by the spending category j (household consumption, government consumption, and fixed investment) by a_j. Absorption increases by $(1-\sigma)\widetilde{R}_t$, and the increase is split across the spending categories in the proportions a_j. I expect the share saved to be high at

the onset of the windfall, because governments have prepared their budgets in advance. The saved share should decline as the government and the private sector incorporate the windfall into their budgets, transfer schemes and investment plans and start to spend it. The share of the increase in absorption spent on investment likely depends on fiscal institutions as well, as evidence suggests that commodity-exporting countries tend to run procyclical fiscal policies (Céspedes and Velasco 2014).

2.2 From spending to emissions

Emissions are the sum of the emission intensity of each sector multiplied by the activity in that sector, $E_t = \sum_s e_s Q_{s,t}$. Each dollar of absorption in each spending category affects the sectoral activities Q through the input requirements of the spending categories net of imports. The total carbon content of the windfall revenues is therefore

$$\kappa = \frac{\sum_h \Delta E_{t+h}}{\sum_h \Delta R_{t+h}} = (1-\sigma) \sum_j a_j \, c_j, \qquad c_j = \sum_s e_s \, \frac{\partial Q_s}{\partial A_j} \tag{1}$$

where A_j is spending of type j and c_j is the CO_2 emitted inside the country's border per dollar of such spending. Because a fraction of the rent will be saved or spent abroad, κ will be lower than the average carbon intensity of GDP, unless the absorption is disproportionately carbon intensive. The composition of κ will follow the composition of absorption. Household consumption will increase fuel consumption in road transport and energy consumption in buildings. Spending on investment will lead to increased production of the inputs to construction, such as cement, steel, and other construction materials. All spending will lead to increased demand for electricity. As rent windfalls are generated by price changes, while volumes are driven by geology, capacity, and, for OPEC members, quotas, I expect rent windfalls to have no effect on the emissions related to extraction.

2.3 Capacity and utilisation

Carbon-intensive industries have putty-clay capital (Atkeson and Kehoe 1999). The emission intensity of these industries is a characteristic of the vintage of the capital, such as cement kilns and blast furnaces. The firms can choose the input proportions when they build a new plant, but after the plant is constructed the input proportions are fixed. In addition, these industries have costly and irreversible capacity, which they usually do not use fully. The demand for construction will increase the output of some of these industries, for instance clinker production in cement kilns, as long as there is slack capacity available to meet the demand. I denote the output of cement by Q_t, public and private investment by I_t^g and

I_t^p, the amount of cement used per unit of investment by θ, the installed capacity of clinker production by K_t and the maximum utilisation rate by $u^{\max}$. The output and emissions of the cement industry are then

$$Q_t = \min\left\{\theta\left(I_t^g + I_t^p\right),\ u^{\max} K_t\right\}, \qquad E_t^{cem} = e\, Q_t \tag{2}$$

where e is the CO_2 emitted per tonne of cement. A windfall therefore raises Q_t through the utilisation of installed capacity, while investment in capacity will be based on the expected long-run demand and will not increase in response to temporary unexpected price increases. Under this structure, short-run changes in energy prices should barely move the output of existing plants, which is what the evidence shows (Appendix Table A16).

2.4 Asymmetry and predictions

A positive rent windfall will lead to an increase in the stock of durable goods and structures, such as cars, buildings, home appliances, roads, and public facilities, which remain in use long after the revenue is spent. A negative rent windfall, on the other hand, will only lead to lower new expenditures, including lower public investment. The stock of durable goods will not decline in a downturn, because public projects begun during a boom are usually finished. The emissions related to the stock of durable goods will therefore respond more to positive windfalls than to negative windfalls, and together with the capacity and absorption arguments above, the framework gives five predictions:

P1. The windfall is first saved and then absorbed into consumption, investment, and imports.

P2. Emissions will increase in the sectors serving the absorbed demand, but the extraction sector's emissions will not be affected. In total, the emissions per dollar of rent will be lower than the average carbon intensity of GDP.

P3. Emissions in the sectors related to construction will increase through an increase in the utilisation rate of the existing capacity.

P4. Emissions related to the stock of durables will increase more in booms than they will decrease in busts of the same size.

P5. Institutions that force saving, or that pass rents to other spending categories before they enter the investment budget, will reduce the public investment response.

If rent-rich economies simply rode global booms, then controlling for the global cycle would make the response vanish, and OPEC supply news would have no effect. If rent windfalls only increase the scale of economic activity, then all sectors will increase in proportion to GDP.

3 Data

3.1 Emissions

I obtain fossil CO_2 emissions, both total and according to the categories of the 2006 guidelines of the Intergovernmental Panel on Climate Change (IPCC), from EDGAR 2026 (Crippa et al. 2026), which reports annual emissions from 1970 to 2025. I aggregate the categories into power (1.A.1.a), refining and other energy industries (1.A.1.b and 1.A.1.c), manufacturing combustion (1.A.2), road transport (1.A.3.b), other transport (the other sectors in 1.A.3), buildings (1.A.4), extraction and fugitive emissions (1.B), production of construction materials (process emissions from production of cement, lime, glass, and other carbonates as well as metals, that is, sectors 2.A and 2.C), production of chemicals and other non-energy products (2.B and 2.D), and a residual sector. These ten sectors sum to total fossil CO_2 emissions. For many developing countries, EDGAR emissions are constructed from emissions-relevant activity statistics, so I compare the results with two independent sources. Emissions from cement production come from the Global Carbon Project cement database (release 260526; see Andrew 2019 for details). For each country-year, these data provide emissions and production, both in tonnes, and a flag that indicates the source of the data. I do not include interpolated years in my analysis, because interpolation could create spurious persistence. Total energy-related CO_2 emissions come from the international energy statistics of the U.S. Energy Information Administration (2026), which are based on measured consumption of individual fuel products and from which the EIA calculates energy CO_2. I restrict the sample to country-years for which the previous year's total fossil CO_2 emissions are at least one million tonnes. This removes a number of micro economies for which EDGAR reports significant breaks in emission levels. The resulting sample comprises 163 countries.[1]

[1]For example, the EDGAR series of Greenland rises by a factor of more than 400 in 2004, and that of Equatorial Guinea by a factor of 15 in 1994. Over the full sample of 195 countries, the cement estimate at five years is 0.95, compared with 0.91 for the analysis sample (Section 5.4). Such breaks make the response of log total CO_2 imprecise across all 195 countries (0.05, standard error 0.19). When emissions are measured as their change relative to the previous level, a measure on which breaks in the series of small emitters have little effect, the response is 0.33 (0.09), the same as in Section 5.2.

3.2 Fossil rents and the windfall shock

For each country and each of the three fossil fuels (oil, natural gas, and coal), I obtain the ratio of rents to GDP from the World Development Indicators (World Bank 2026a). I calculate exposure $s_{cf,t}$ according to (3) as the average of this ratio over $t-10$ to $t-6$. I calculate world prices as the annual mean of monthly prices from the World Bank (2026b), deflated by the US consumer price index (Federal Reserve Bank of St. Louis 2026). I estimate the coefficients ϕ_1 and ϕ_2 of an autoregression of order two, AR(2), for the log real world price of each fuel using only years up to $t-1$, and calculate the price innovation u_{ft} as the difference between the realised log price and the forecast.[2] I then calculate the windfall according to

$$W_{ct} = \sum_{f \in \{\text{oil, gas, coal}\}} s_{cf,t}\, u_{ft}, \qquad s_{cf,t} = \frac{1}{5} \sum_{k=6}^{10} \frac{R_{cf,t-k}}{Y_{c,t-k}}. \tag{3}$$

As $s_{cf,t}$ is a share and u_{ft} is a log difference, W_{ct} is an approximation of the windfall $\widetilde{R}_t$ of Section 2.1 as a share of GDP. The standard deviation of W_{ct} is 0.036 across all country-years in the sample (Table 1), and 0.086 when only including country-years with exposure of at least 0.10; the latter is the typical windfall I use to interpret magnitudes.[3] A unit shock corresponds to a country receiving unexpected fossil fuel revenues equal to its GDP in a given year. As an indication of the magnitude, the shock for Saudi Arabia was 0.091 in 2005 and -0.288 in 2015.

3.3 Oil supply news

In the supply-driven variant, I use monthly oil supply news shocks from Känzig (2021), vintage 2025M12, which are identified from changes in oil futures prices within a narrow window around OPEC announcements and are thus not driven by global demand shocks. Since I use annual averages of oil prices, I aggregate these monthly news shocks to annual ones based on the share of each year affected by a shock in month m, namely $(13-m)/12$ for the year in which the shock happens and the remaining $(m-1)/12$ for the following year. I then use oil exposure times these aggregated news shocks as an external instrument for the windfall shock in the local projection regressions, following Stock and Watson (2018). I estimate these local projections for the 147 non-OPEC countries in my sample, since the news shocks are based on OPEC's announcements, which are decisions of its own members.

[2] Each year, the autoregression is estimated again on all preceding years, so that every innovation relies only on information from before the shock. Forecasts begin when a price history of twelve years is available, that is, from 1975 for oil, 1982 for coal and 1989 for gas; until then, the shock has no coal or gas component.

[3] To express magnitudes for a typical windfall, I multiply each coefficient by 0.086, reading log points as percentages.

The first-stage F statistic, clustered by country and year, ranges from 7 to 24 for the different outcomes (Panel B of Tables 2 and 3). When I use the simpler aggregate of oil news shocks (that is, the annual sum, ignoring the month in which the shock happens within the year), the first stage is much weaker (Table A7).

3.4 Government revenue, national accounts, capacity and institutions

Data on government revenues, including resource revenues and other types of revenues, come from the UNU-WIDER Government Revenue Dataset 2025 (UNU-WIDER 2025; McNabb 2017), and data on real GDP, household and government consumption, gross fixed capital formation, exports, imports and the current account balance come from the World Bank's World Development Indicators (World Bank 2026a). Public and private investment come from the IMF's Investment and Capital Stock Dataset, which is only available until 2019 (International Monetary Fund 2021). Data on the capacity, start year and coordinates of cement and iron and steel plants come from the Global Energy Monitor's cement and iron and steel trackers (Global Energy Monitor 2026a,b). These data include plants that were closed in recent years but not those closed earlier. I obtain data on built-up surface from the Global Human Settlement Layer (Pesaresi 2023; Pesaresi et al. 2024). Data on countries' fiscal rules are from the IMF's Fiscal Rules Dataset (Alonso et al. 2025). I measure countries' regime type by the Polity2 score from the Polity5 project (Marshall and Gurr 2020), which I source from Martinez-Alvarez et al. (2022), using their replication data. I measure fuel underpricing using the data of Ross et al. (2017) on the net tax on gasoline, which I extend to 2023 using the World Bank's Global Fuel Prices Database (World Bank 2026c) and replication data from Mahdavi et al. (2025). Data on deaths in conflicts are taken from the Uppsala Conflict Data Program (UCDP; Davies et al. 2026). I provide an overview of all data sources in Table A1.

Average fossil rents from 2005 to 2014 are largest in the Gulf region, North and Central Africa, around the Caspian Sea and in Russia, and these regions also contain many of the 2,481 currently operating integrated cement plants that the Global Energy Monitor's cement tracker lists outside China (Figure 1a). Fossil fuel prices in real terms (deflated by the US consumer price index) exhibit long swings but are subject to considerable annual surprises; the boom from 2003 to 2014 is particularly visible (Figure 1b). The annual aggregates of the oil supply news shocks and the oil price innovations, measured in real time, are correlated (0.31 between 1976 and 2024), but far from perfectly (Figure 1c).

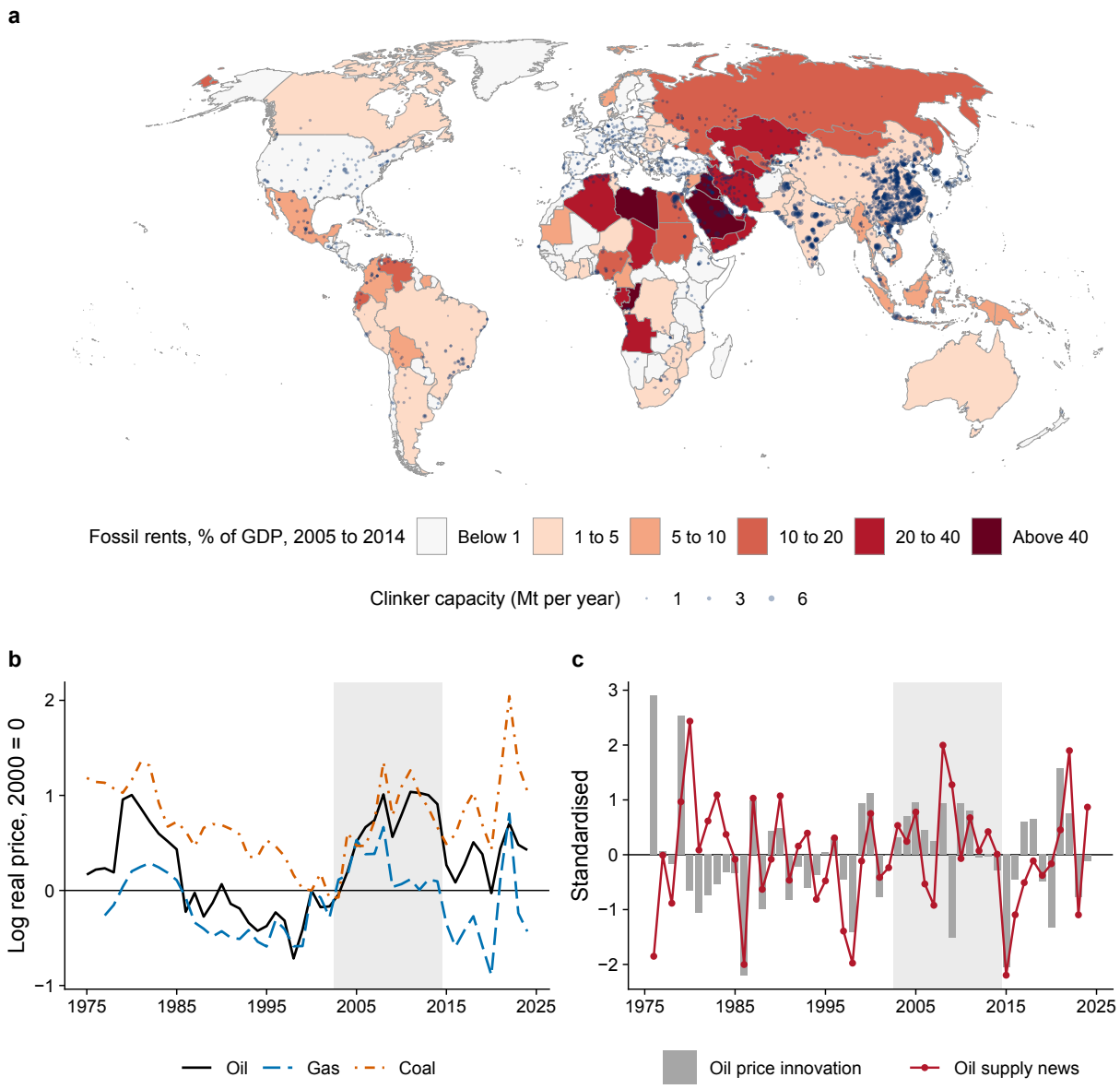


Figure 1: Fossil rent exposure and integrated cement plants (a), real fossil fuel prices (b), and oil price innovations with oil supply news (c). Shaded years: the 2003 to 2014 commodity boom

Table 1: Summary statistics

Variable	N	Mean	SD	Min	Max
Fossil rents, share of GDP ($t-10$ to $t-6$)	5,814	0.057	0.116	0.000	0.829
Windfall shock	5,814	−0.000	0.036	−0.402	0.272
Supply-news instrument (outside OPEC)	5,147	−0.002	0.096	−1.787	1.048
Total fossil CO_2 (Mt)	5,814	196.839	813.703	0.648	13,125.071
Road transport CO_2 (Mt)	5,814	32.712	125.864	0.019	1,544.553
Buildings CO_2 (Mt)	5,814	21.534	71.685	0.000	663.318
Power CO_2 (Mt)	5,814	72.390	342.824	0.000	6,645.972
Cement process CO_2 (Mt)	5,646	6.928	41.867	0.000	821.600
Construction materials process CO_2 (Mt)	5,814	10.615	66.621	0.000	1,328.147
Cement production (Mt)	5,599	18.238	127.821	0.000	2,492.071
Public investment (billion 2017 PPP USD)	4,660	31.889	145.579	0.000	3,124.310
Real GDP (billion 2015 USD)	5,749	413.581	1,564.109	0.353	22,731.468

Notes: The table summarises the country-year panel used in the main estimates. The sample is country-years with total fossil CO_2 of at least 1 Mt in $t-1$, shocks from 1981 to 2024. Rents are World Development Indicators oil, gas and coal rents averaged over $t-10$ to $t-6$; the windfall shock sums each fuel's rents share times the real-time innovation in its real world price; the supply-news instrument is oil rents times the Känzig (2021) oil supply news shocks aggregated to years by the share of each year that a monthly shock affects. Emissions are EDGAR 2026, cement production the Global Carbon Project, public investment the IMF Investment and Capital Stock Dataset (to 2019). No significance tests are reported.

4 Empirical strategy

4.1 Local projections

I estimate the following local projection for each horizon $h = 0, \ldots, 8$:

$$y_{c,t+h} - y_{c,t-1} = \beta_h W_{ct} + \sum_{j=1}^{2} \gamma_{hj} W_{c,t-j} + \sum_{j=1}^{2} \rho_{hj} \Delta y_{c,t-j} + \alpha_{ch} + \delta_{th} + \varepsilon_{c,t+h} \tag{4}$$

where y refers to the logarithm of the outcome variable, or to its level relative to GDP when the variable is revenue or the current account, and α and δ are country and year fixed effects. Leads and lags are included based on calendar years, drawn from the complete country-year panel, and the sample is not restricted in years other than the year of the shock. Thus, a country may have an observation for the outcome variable five years after a shock even if its emissions are below the threshold in those years. Including lagged shocks controls for the impact of past windfalls, and including lagged growth of the outcome controls for momentum and ensures that inference is robust to the persistence of the outcome variable (Montiel Olea and Plagborg-Møller 2021). With these controls, local projections estimate the same impulse response functions as a vector autoregression in population (Plagborg-Møller and Wolf 2021). I plot the estimated coefficients, which are cumulative responses, from the fourth year before the shock to the eighth year after it, along with 90 and 95 per cent confidence intervals, and

shade the years before the shock. Standard errors are clustered by country and year because all countries are exposed to the same price innovations in a given year. I implement the eigenvalue method of Cameron et al. (2011) when the two-way clustered variance-covariance matrix is not positive semi-definite, and each event-study figure reports the coefficients for $k = 2, 3, 4$ years before the shock.

4.2 Carbon content of windfall revenue

To express the footprint per dollar, I estimate

$$\sum_{h=0}^{8} \frac{E_{c,t+h} - E_{c,t-1}}{Y_{c,t-1}} = \kappa \sum_{h=0}^{8} \frac{R_{c,t+h} - R_{c,t-1}}{Y_{c,t-1}} + \mathbf{x}'_{ct}\lambda + \alpha_c + \delta_t + \nu_{ct} \tag{5}$$

I use two-stage least squares, instrumenting cumulative rent revenue with the windfall shock W_{ct}. Emissions are measured in tonnes and revenue in current dollars, and each is expressed relative to the level of GDP prior to the windfall shock. Thus, κ measures the additional CO_2 produced within a country's borders per extra dollar of fossil fuel revenue over the nine years starting in the year of the shock. I refer to the extra revenue as "windfall revenue" and to κ as the carbon content of windfall revenue. The controls, $\mathbf{x}$, consist of two lags of the shock and the changes in emissions and revenue over the year before the shock. I also run this regression for each sector to obtain the carbon content by sector. Given that I have a single instrument, the Anderson-Rubin test of the null hypothesis that $\kappa = 0$ is equivalent to testing the reduced form. I invert this test to obtain a confidence set for κ that is robust to weak instruments.

This ratio is interpreted as the carbon content of windfall revenue so long as price surprises act on emissions solely through their impact on revenue. To the extent that domestic fuel prices track world prices, windfall price surprises also raise the cost of domestic fuel, which would bias my estimate κ towards zero. Furthermore, if economies with large exposure to fossil fuel prices respond differently to the concurrent upturn in global economic activity, my estimate κ would also pick up this differential response. These channels do not appear to drive the estimate (Table A5). In economies that underpriced fuel in the years before the shock, so that the domestic price of fuel is not linked to the world price, I estimate a carbon content of 0.36 kg. Adding interactions of exposure with proxies for the global cycle raises the full-sample estimate to 0.59 kg. I also use the price surprises of oil and of gas separately as the instrument, since they affect domestic fuel markets and global demand differently; the carbon content is then 0.50 kg and 0.74 kg, respectively. A test of the hypothesis that the carbon content is identical across fossil fuels has a p-value of 0.08, so I cannot reject this

null hypothesis at the 5 per cent level. Without the exclusion restriction, κ still represents the impact of a surprise in fossil fuel prices on emissions per dollar of revenue generated, an average that gives more weight to the economies that depend most on rents.

4.3 Identifying assumption and inference

Year effects absorb the direct effect of world prices and every other common shock, so the comparison is between countries with larger and smaller predetermined rents in years with larger and smaller surprises. As Goldsmith-Pinkham et al. (2020) and Borusyak et al. (2025) note, the identifying assumption is violated if the exposure variable is correlated with differential shocks to emissions and economic activity in years with large surprises. The exposure variable is constructed six to ten years prior to the shock and the innovations are forecast errors, so neither reacts to the country's current conditions. The main potential violations of the identifying assumption in my setting include differential exposure to the global cycle, large exporters setting prices, and conflict or sanctions coinciding with high prices. In Section 5.5, I address each of these concerns by interacting exposure with proxies for the global cycle, including region-by-year effects, dropping OPEC and Russian observations, controlling for conflict and using the oil supply news as an instrument. As the identifying variation stems from a small number of price surprises, I also use randomisation inference, calculate standard errors from an equivalent regression at the level of the fuel-year surprises, and report leave-one-year-out estimates and estimates that drop each major price episode in turn (Adão et al. 2019; Borusyak et al. 2022).

5 Empirical results

5.1 Macroeconomic transmission

The dependent variables are measured in log points unless otherwise noted; rent revenue and the current account are expressed as changes relative to the country's GDP before the shock. A unit windfall shock increases fossil rent revenue by 0.81 relative to GDP on impact (Table 2; Figure 2). The current account increases by 0.64, implying that the bulk of the revenue is saved abroad in the year of the windfall, while government consumption (0.24), fixed investment (0.34), public investment (0.54), and imports (0.36, Table A2) also start to increase. Two years after the shock, the response of the current account has subsided to 0.17 and is not statistically different from zero, while the revenue increase remains high (0.58), which implies that the windfall has been absorbed. After two years, household consumption has increased by 0.31, government consumption by 0.77, fixed investment by

1.20 and imports by 1.03. After five years, the corresponding responses are 0.45, 0.86, 1.45 and 0.99, and public investment peaks three years after the shock at 1.97. Real GDP has increased by 0.37 five years after the shock. Exports do not respond to the shock (Table A2). None of the pre-period coefficients in Panel A of Table 2 is statistically significant.

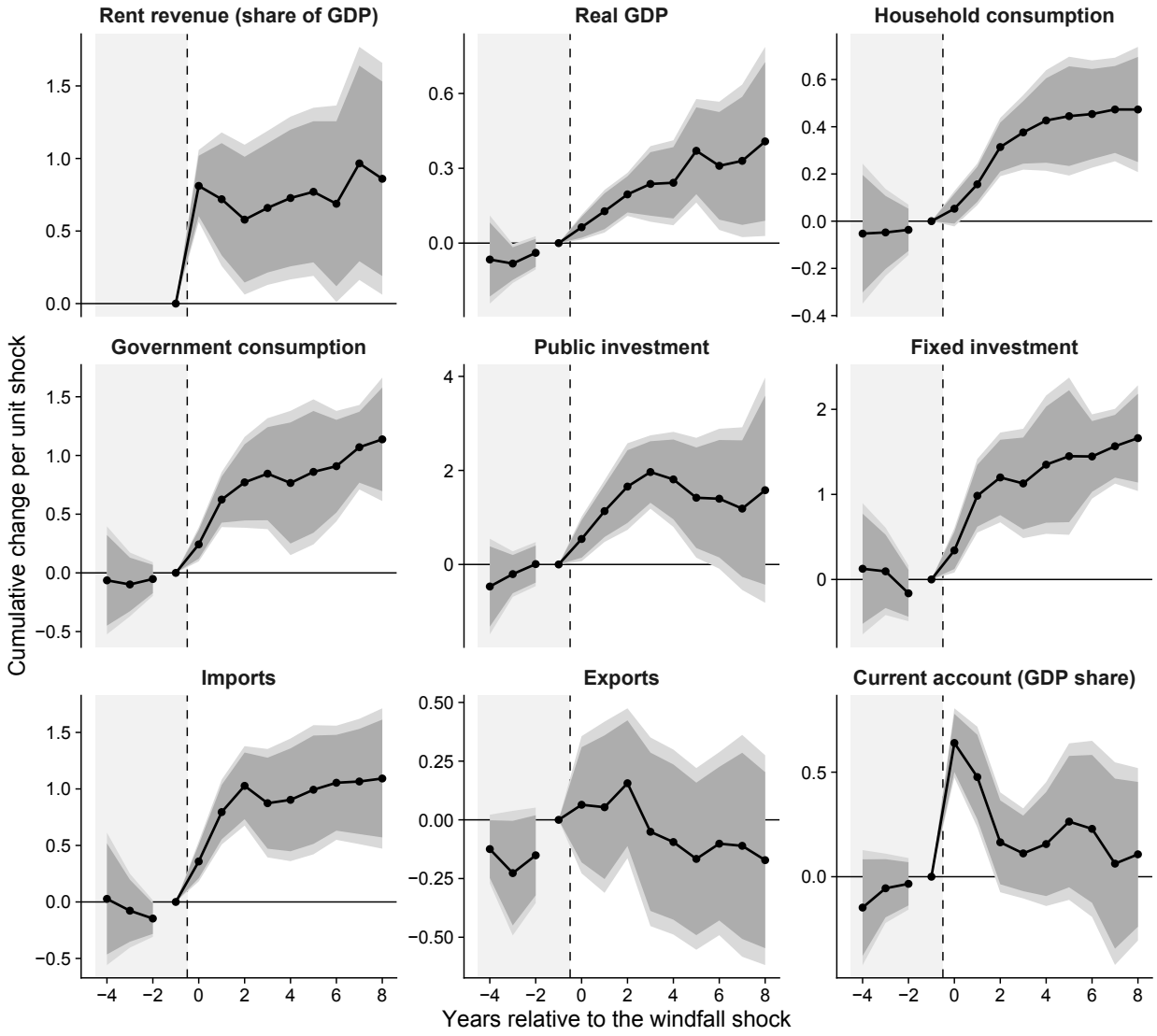


Figure 2: Macroeconomic responses to a fossil windfall. Dark and light bands: 90 and 95 per cent confidence intervals; shaded area: pre-period coefficients

The windfalls are captured in the government budget. I exploit information from UNU-WIDER on government resource revenue, which is available for 75 resource-producing economies. Per unit shock, the increase in government resource revenue is 0.56 on impact, between 0.74 and 0.80 in the following three years, and 0.97 after five years, while the corresponding increase in fossil rent revenue in these country-years is between 0.60 and

Table 2: Macroeconomic transmission of a fossil windfall

Horizon	Rent revenue	Real GDP	Household consumption	Government consumption	Public investment	Fixed investment	Current account
Panel A. Windfall shock (price innovations)							
Pre-period, $h = -2$	-0.032	-0.039	-0.037	-0.052	0.008	-0.163	-0.034
	(0.075)	(0.034)	(0.054)	(0.073)	(0.239)	(0.167)	(0.063)
$h = 0$	0.811^{***}	0.064^{**}	0.053	0.243^{***}	0.541^{**}	0.342^{**}	0.640^{***}
	(0.126)	(0.025)	(0.038)	(0.074)	(0.242)	(0.133)	(0.085)
$h = 2$	0.578^{**}	0.195^{***}	0.314^{***}	0.771^{***}	1.656^{***}	1.199^{***}	0.165
	(0.263)	(0.044)	(0.063)	(0.197)	(0.470)	(0.269)	(0.122)
$h = 3$	0.659^{**}	0.237^{***}	0.376^{***}	0.845^{***}	1.966^{***}	1.129^{***}	0.111
	(0.271)	(0.077)	(0.080)	(0.241)	(0.398)	(0.328)	(0.110)
$h = 5$	0.770^{**}	0.370^{***}	0.445^{***}	0.860^{***}	1.419^{**}	1.449^{***}	0.264
	(0.296)	(0.106)	(0.128)	(0.315)	(0.650)	(0.471)	(0.191)
$h = 8$	0.860^{**}	0.408^{**}	0.473^{***}	1.137^{***}	1.579	1.662^{***}	0.107
	(0.407)	(0.193)	(0.135)	(0.268)	(1.222)	(0.317)	(0.210)
Observations ($h = 5$)	4,815	4,693	3,607	3,593	3,648	3,552	4,012
Panel B. Windfall shock instrumented by oil supply news, outside OPEC							
Pre-period, $h = -2$	-0.110	-0.037	-0.184	-0.422^{**}	-0.546	-0.833^{**}	0.060
	(0.078)	(0.060)	(0.162)	(0.166)	(0.396)	(0.348)	(0.043)
$h = 0$	0.660^{***}	0.230^{***}	0.352^{**}	0.394^{**}	1.474^{**}	1.262^{***}	0.635^{***}
	(0.184)	(0.061)	(0.137)	(0.178)	(0.712)	(0.032)	(0.084)
$h = 2$	0.101	0.439^{***}	0.548^{**}	1.159^{***}	2.622^{**}	1.048^{*}	0.069
	(0.326)	(0.102)	(0.251)	(0.181)	(0.972)	(0.518)	(0.217)
$h = 3$	0.399	0.424^{***}	0.677^{**}	1.180^{***}	2.006^{***}	0.551	-0.010
	(0.266)	(0.117)	(0.288)	(0.433)	(0.601)	(0.939)	(0.239)
$h = 5$	0.214	0.356^{**}	0.726^{*}	1.403^{*}	0.013	1.567^{*}	-0.142
	(0.410)	(0.167)	(0.416)	(0.788)	(1.281)	(0.884)	(0.364)
$h = 8$	-0.408	0.240	0.277	0.101	-1.878	0.314	-0.597
	(0.925)	(0.282)	(0.171)	(1.194)	(3.007)	(1.399)	(0.398)
First-stage F ($h = 5$)	17.5	17.4	7.1	7.1	9.6	7.1	20.9
Observations ($h = 5$)	4,276	4,136	3,285	3,271	3,214	3,237	3,643

Notes: Each coefficient is the cumulative change in the outcome from $t-1$ to $t+h$ per unit shock in t, from a separate local projection; outcomes are in logs except rent revenue and the current account. Rent revenue is the change in World Development Indicators oil, gas and coal rents from $t-1$ to $t+h$ divided by GDP in $t-1$, so a coefficient of one means extra rents equal to the windfall; its regressions control for two lags of the shock and the change in rents before the outcome window, and drop country-years in which either change exceeds GDP. The current account is the change in the current account balance as a share of GDP, so that a coefficient of one in Panel A means the whole windfall is saved abroad. Panel B instruments the windfall shock with oil rents times the Känzig (2021) oil supply news shocks, aggregated to years as in Table 1, in countries outside OPEC (local projections with an external instrument), so its coefficients are on the scale of Panel A; the first-stage F statistic is the two-way clustered Wald statistic of the excluded instrument at $h = 5$. The sample is country-years with total fossil CO_2 of at least 1 Mt in $t-1$, shocks from 1981 to 2024. Estimation is by OLS local projections with country and year fixed effects and two lags of the shock and of the outcome change; standard errors in parentheses are two-way clustered by country and year. Real aggregates are World Development Indicators series in constant 2015 dollars; public investment is from the IMF Investment and Capital Stock Dataset (to 2019). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

0.80. Government non-resource revenue does not respond much to the shock (Table A3). Regressing the cumulative increase in government resource revenue on the cumulative increase in fossil rents, with the windfall shock as the instrument, I find that 0.76 of fossil rents are captured in government resource revenue over three years and 1.01 over five years. These fractions are estimated only for the countries for which data are available, and they are not a mechanical consequence of how the shock is constructed. Under the supply news instrument, only the revenue response on impact is statistically significant. In this sample, the spending responses reflect countries whose budgets have just received a windfall.

A typical shock of 0.086 implies an increase in revenue of around 7 per cent of GDP in the first year and an increase in the current account of around 5.5 per cent of GDP. After five years, household consumption is 3.8 per cent higher, government consumption is 7.4 per cent higher, fixed investment is 12.5 per cent higher, and imports are 8.5 per cent higher. Because the revenue comes in during a fiscal year for which spending has already been set, it is first saved and then incorporated into government budgets in subsequent years. The persistent increases in consumption and investment are in line with the procyclical fiscal policy observed in commodity-exporting countries (Céspedes and Velasco 2014). The large increase in imports, which take up much of the extra spending, is one of the reasons why a windfall dollar has a lower carbon content than an average dollar of GDP.

5.2 The carbon footprint of a windfall

5.2.1 Emissions by sector

Total fossil CO_2 emissions increase significantly on impact, by 0.10 log points (Table 3, Panel A; Figure 3), by 0.26 after two years, by 0.27 after five years, and by 0.25 even after eight years. CO_2 emissions from fossil fuel combustion in road transport increase by 0.24 on impact and 0.76 after five years. Emissions from the buildings sector increase by 0.21 on impact and 0.92 after five years. Emissions from fossil fuel combustion in manufacturing and from construction materials also increase, by 0.53 and 0.73, respectively, after five years. Emissions from the power sector increase from the third year onwards, but the point estimates, which range from 0.1 to 0.3, are imprecise in logs, and there is no response from the extraction sector. The pre-period is flat for total CO_2 emissions, road transport and buildings, the series on which the footprint rests. In contrast, the coefficients for the third and fourth year before the windfall are significant in the chemicals sector, and the coefficient for the fourth year is marginally significant for combustion in manufacturing, so I place no weight on the timing of the responses in these two sectors.

If the windfall increased the scale of economic activity proportionally in all sectors, the

Table 3: Emissions by sector after a fossil windfall

Horizon	Total fossil CO_2	Road transport	Buildings	Power	Manufacturing combustion	Construction materials
Panel A. Windfall shock (price innovations)						
Pre-period, $h = -2$	-0.035	0.000	0.012	-0.068	-0.094	-0.014
	(0.038)	(0.062)	(0.084)	(0.129)	(0.123)	(0.063)
$h = 0$	0.096^{**}	0.235^{**}	0.213^{**}	-0.138	0.149^{***}	0.190
	(0.042)	(0.087)	(0.094)	(0.115)	(0.045)	(0.215)
$h = 2$	0.263^{***}	0.499^{***}	0.535^{***}	0.005	0.295^{**}	0.223
	(0.073)	(0.154)	(0.190)	(0.199)	(0.143)	(0.203)
$h = 3$	0.236^{**}	0.509^{***}	0.666^{***}	0.120	0.415^{**}	0.456^{*}
	(0.101)	(0.177)	(0.227)	(0.231)	(0.200)	(0.232)
$h = 5$	0.265^{***}	0.760^{***}	0.916^{***}	0.293	0.530^{*}	0.733^{**}
	(0.073)	(0.223)	(0.302)	(0.292)	(0.263)	(0.356)
$h = 8$	0.253^{*}	0.773^{**}	0.997^{**}	0.280	0.485	0.618
	(0.129)	(0.298)	(0.416)	(0.278)	(0.507)	(0.453)
Observations ($h = 5$)	4,743	4,743	4,705	4,648	4,677	4,743
Panel B. Windfall shock instrumented by oil supply news, outside OPEC						
Pre-period, $h = -2$	-0.229^{**}	-0.097	-0.626	-0.266	-0.150	0.229
	(0.107)	(0.098)	(0.548)	(0.246)	(0.311)	(0.405)
$h = 0$	0.326^{*}	0.642^{**}	0.402	0.163	0.964^{***}	0.206
	(0.184)	(0.299)	(0.343)	(0.300)	(0.313)	(0.234)
$h = 2$	0.131	0.685^{**}	1.104^{*}	0.247	0.963^{*}	0.235
	(0.199)	(0.261)	(0.602)	(0.290)	(0.547)	(0.458)
$h = 3$	0.135	0.642^{*}	1.239^{**}	0.065	1.299^{**}	0.471
	(0.176)	(0.318)	(0.612)	(0.330)	(0.568)	(0.622)
$h = 5$	-0.147	0.609	1.944^{***}	-0.085	0.107	0.187
	(0.160)	(0.363)	(0.647)	(0.396)	(0.406)	(0.836)
$h = 8$	-0.347	0.057	1.540	0.176	0.041	-0.611
	(0.397)	(0.491)	(1.047)	(0.332)	(0.894)	(0.879)
First-stage F ($h = 5$)	18.8	19.0	23.8	18.9	18.9	18.9
Observations ($h = 5$)	4,186	4,186	4,148	4,103	4,126	4,186

Notes: Each coefficient is the cumulative log change in the sector's fossil CO_2 (EDGAR 2026, IPCC 2006 categories) from $t - 1$ to $t + h$ per unit shock in t, from a separate local projection. Road transport is 1.A.3.b, buildings 1.A.4, power 1.A.1.a, manufacturing combustion 1.A.2, and construction materials the process emissions of cement, lime, glass, other carbonates and metals (2.A and 2.C). Panel B is estimated as in Table 2. Sample, estimation and significance levels as in Table 2.

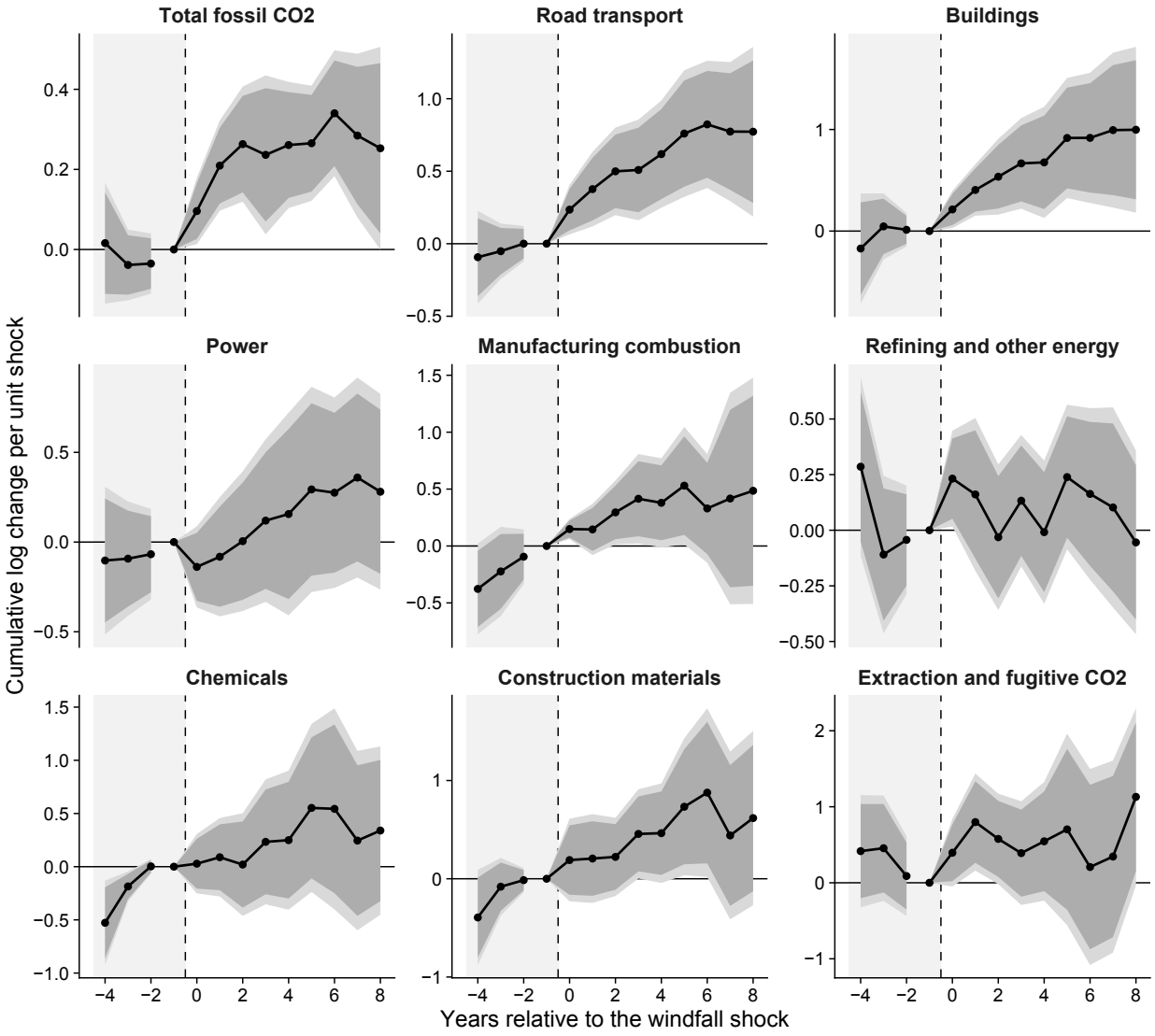


Figure 3: Emissions by sector after a fossil windfall. Bands and shading as in Figure 2

response of every sector would equal the response of GDP, which is 0.37 after five years (Table 2). However, the responses of emissions from road transport, buildings, manufacturing combustion and construction materials are about one and a half to two and a half times as large, whereas in the extraction sector there is no increase. The results imply that the increase in activity induced by the windfall is disproportionately directed towards the energy-intensive parts of consumption and construction.

5.2.2 Sector contributions and carbon content

The log specification estimates elasticities, so high elasticities for small sectors do not necessarily imply a large contribution to the increase in carbon emissions. The contributions, stacked by sector group in Figure 4, come from regressions of the change in each sector's carbon emissions, divided by total carbon emissions before the shock, with the windfall shock as the explanatory variable and the same sample and regressors for all sectors, so that the coefficients add up to the total. After five years, a unit shock increases total carbon emissions by 0.33 of their pre-shock level. Road transport accounts for the largest increase (0.10), followed by power (0.09), buildings (0.07), refining (0.06), combustion in manufacturing (0.04), chemicals (0.03) and construction materials (0.02) (Table 4, column 1; Figure 4a).

Each dollar of windfall revenue increases domestic carbon emissions by 0.52 kg over nine years (Table 4, column 2). This estimate has a standard error of 0.18 and a first-stage F statistic of 11.5, and its 95 per cent Anderson-Rubin set, [0.20, 0.89] kg, is similar to the Wald interval (Table A5). The average carbon intensity of GDP in high-rent economies (rents of at least 5 per cent of GDP), with exposure times GDP as weights, is 1.04 kg of CO_2 per dollar (Table 4). Hence each dollar of the windfall carries about half as much carbon as an average dollar of GDP in a resource-rich economy, which is in line with the prediction of the framework, since part of the windfall is saved abroad or spent on imports. Roughly one quarter of this carbon comes from road transport (0.14 kg per dollar of revenue), one quarter from the power sector (0.12 kg), one tenth from the buildings sector, 6 per cent from combustion in manufacturing and 7 per cent from the production of construction materials, mostly cement (Figure 4b). Compared with the emission shares of the sectors in economies whose rents are at least 10 per cent of GDP (Table 4, column 4), the carbon content of a windfall is tilted towards road transport, buildings and the production of construction materials, while it is tilted away from extraction (17.5 per cent of emissions in these economies against 3 per cent of the carbon content of a windfall). In robustness checks, the estimate ranges between 0.48 and 0.59 kg at different horizons, with different trimming rules, when controlling for the global cycle and when excluding China (Table A4). Excluding OPEC members and Russia leads to a weaker first stage and a wide but bounded Anderson-Rubin

Table 4: Sector contributions to emissions growth and the carbon content of windfall revenue

Sector	Contribution to total CO_2 growth, $h = 5$	Carbon content, kg CO_2 per USD of windfall	Share of carbon content	Share of total CO_2 in rent economies
Road transport	0.103**	0.139***	0.267	0.173
	(0.048)	(0.035)		
Power	0.088	0.117	0.225	0.252
	(0.070)	(0.060)		
Buildings	0.066***	0.054**	0.104	0.086
	(0.021)	(0.021)		
Manufacturing combustion	0.038	0.031	0.060	0.120
	(0.025)	(0.025)		
Refining and other energy	0.058***	0.014	0.027	0.090
	(0.019)	(0.019)		
Construction materials (process)	0.018*	0.036***	0.069	0.035
	(0.009)	(0.013)		
Chemicals and non-energy products	0.031***	0.024**	0.045	0.042
	(0.011)	(0.013)		
Other transport	0.007	0.004	0.008	0.010
	(0.010)	(0.006)		
Extraction and fugitive	−0.018	0.016	0.030	0.175
	(0.053)	(0.051)		
Other	−0.066	0.005	0.009	0.018
	(0.049)	(0.010)		
Total fossil CO_2	0.325***	0.522**	1.000	1.000
	(0.094)	(0.178)		
First-stage F (carbon content)	n.a.	11.5	n.a.	n.a.
Observations	4,624	4,215	n.a.	n.a.

Notes: Column 1 regresses the change in the sector's CO_2 from $t-1$ to $t+5$, divided by total fossil CO_2 in $t-1$, on the windfall shock; every regression uses the same sample and regressors, so the sector coefficients add up to the total, and country-years whose total CO_2 changes by more than 100 per cent are excluded. Column 2 regresses the cumulative extra emissions of the sector over horizons 0 to 8 (tonnes, relative to $t-1$) on the cumulative extra fossil rent revenue over the same horizons (US dollars, relative to $t-1$), both divided by GDP in $t-1$, instrumenting rent revenue with the windfall shock and controlling for two lags of the shock and lagged changes in emissions and rents; the top and bottom one per cent of each cumulative variable are trimmed, and stars in this column are from Anderson-Rubin tests, which in a just-identified model equal the test on the reduced form. All regressions include country and year fixed effects, with standard errors two-way clustered by country and year. Column 3 divides each sector's carbon content by the total; column 4 is the mean share of each sector in total fossil CO_2 in country-years with rents of at least 10 per cent of GDP. In country-years with rents of at least 5 per cent of GDP, the average carbon intensity of GDP, weighted by exposure times GDP, is 1.04 kg of CO_2 per dollar. The sample and significance levels follow Table 2.

set that includes zero (Table A5).

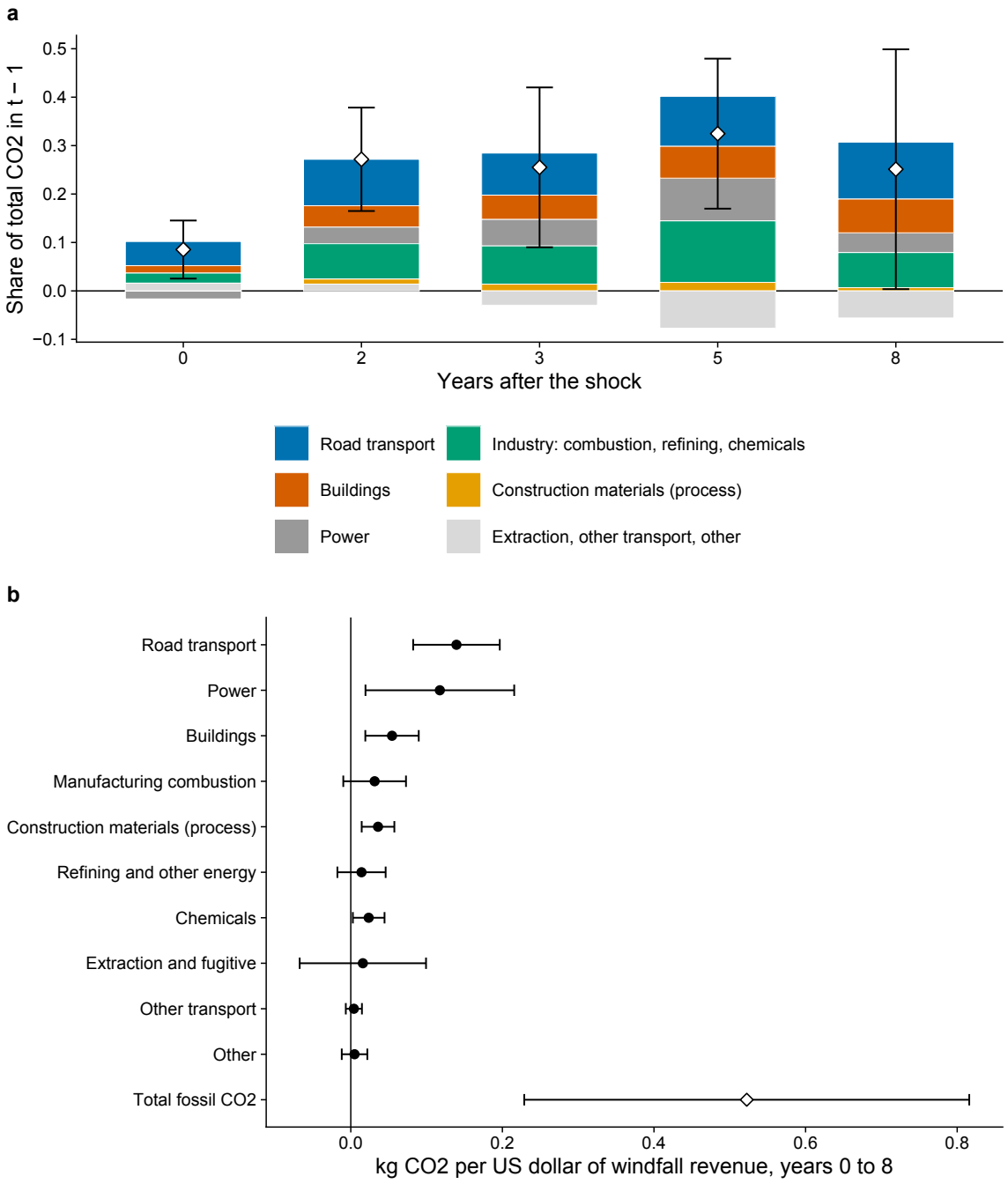


Figure 4: Sector contributions to the growth of total fossil CO_2 (a) and carbon content of windfall revenue by sector (b). Diamonds and whiskers: total with its 90 per cent confidence interval

These conclusions do not hinge on the inclusion of cement emissions. Emissions from construction materials account for less than 10 per cent of the carbon content of a windfall,

and the total carbon content is driven by sectors that are closely linked to household and government expenditure. Cement emissions matter for another reason, which is the subject of Section 5.4.

5.2.3 An independent measure

The EIA energy CO_2 series matches EDGAR very well in levels (correlation of 0.99) but less well in annual changes (correlation of 0.39), so it gives an independent check on the variation on which the estimates rely (Table A6). In the EIA data, energy-related carbon emissions increase by 0.33 after two years and by 0.48 after five years. Measured gasoline and diesel consumption increases by 0.56 and 0.76 at the same horizons. Consumption of all oil products increases by 0.44 and 0.60, while consumption of electricity increases by 0.21 and 0.40. When EIA energy CO_2 replaces EDGAR emissions, the estimated carbon content is 0.51 kg per dollar, with a first-stage F statistic of 11.2 and an Anderson-Rubin test p-value of 0.004 (Table A6), similar to the 0.52 kg obtained with EDGAR (Table 4).

5.3 Supply-driven windfalls

Price innovations are a combination of supply and demand shocks, so Panels B of Tables 2 and 3 repeat the estimation for all 147 non-OPEC countries, using OPEC supply news as the instrument for the windfall shock, and Figure 5 compares these estimates with those based on price surprises for the same 147 countries. The main macroeconomic responses are of similar size to those in the main design: the current account balance improves on impact by 0.64 of GDP, the same as in the main design; real GDP is 0.36 higher after five years, compared with 0.37 in the main design; and government consumption, public investment and household consumption all increase at least as much over the first two years. Similarly, road transport emissions increase by between 0.6 and 0.9 in each of years 0 to 5, while buildings emissions increase by 1.94 at five years and cement emissions by 1.43 at two years (Figure 5). Government resource revenue rises on impact by 0.58, but the responses in later years are estimated imprecisely (Table A3).

Total fossil CO_2 emissions rise on impact by 0.33 and in the following year by 0.45 (Table 3, Panel B; Figure 5), no less than in the main design. The response then drops to about 0.13 in years 2 and 3, and the estimate at each horizon is imprecise. Pooling the horizons, the average increase in total CO_2 emissions over years 0 to 5 is 0.17 ($p = 0.27$) with EDGAR data, 0.26 ($p = 0.01$) with four lags of the shock and the outcome, and 0.48 ($p < 0.001$) with the EIA energy statistics, compared with 0.38 in EDGAR and 0.44 in the EIA data when price surprises are used for the same 147 countries (Table A8). Hence the

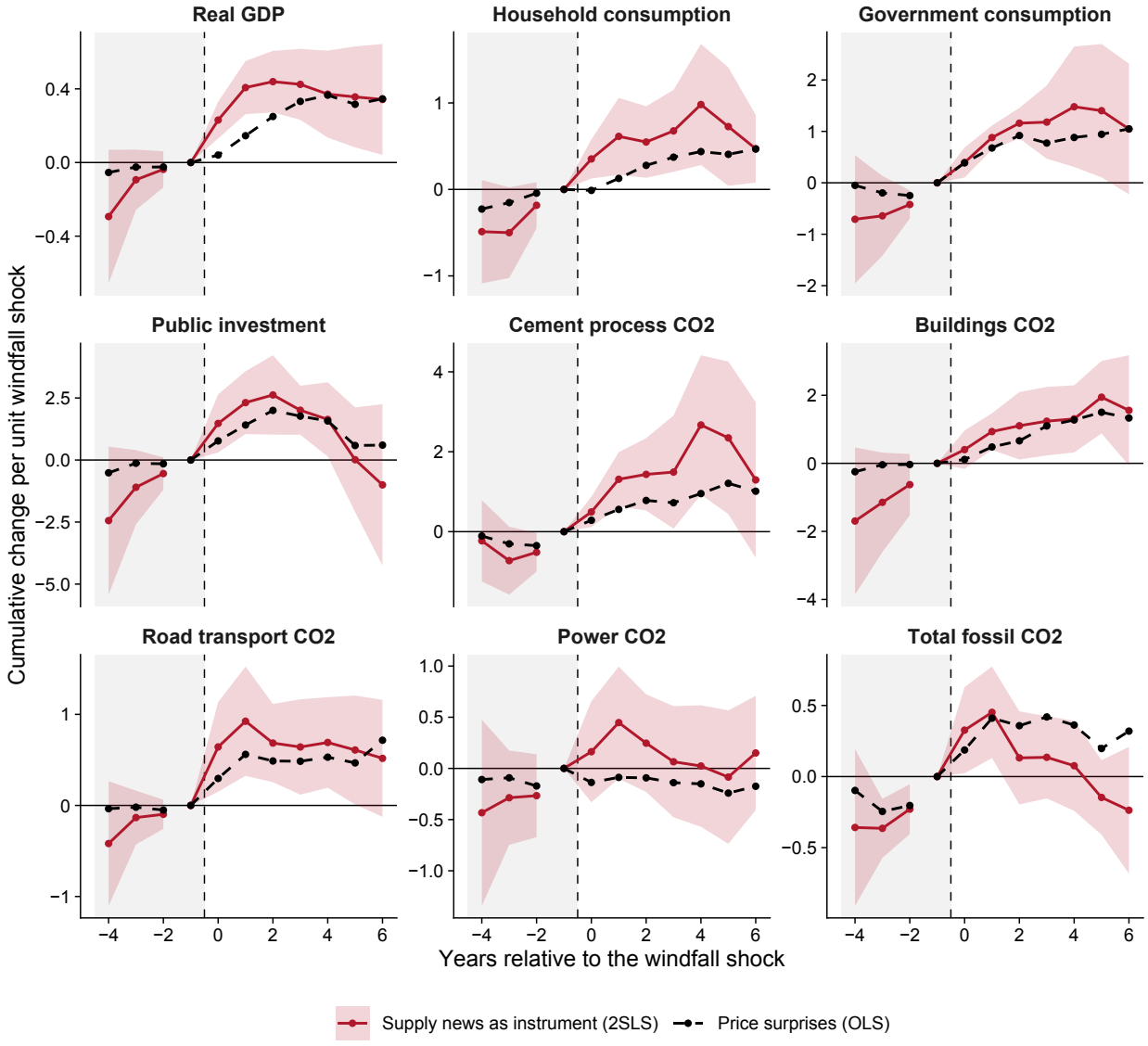


Figure 5: Supply-news and price-surprise responses, countries outside OPEC. Solid lines with 90 per cent bands: supply news as instrument; dashed lines: price surprises

supply-news evidence for total emissions comes from the year of the shock and the year after, the four-lag specification and the independent energy data. Power sector emissions show no significant response to supply news, yet in the main design they account for a large part of the five-year response of total emissions, which partly explains why the response under supply news weakens at longer horizons. A number of pre-period coefficients are negative when supply news is used, particularly for fixed investment, government consumption and total CO_2 emissions; adding four lags removes them for total CO_2, road transport and buildings, while they remain for cement, household consumption and public investment (Table A7). When supply news is the instrument, the first-stage response of cumulative rent revenue is too weak for an estimate of the carbon content per dollar of revenue (Table A5).

5.4 The construction channel

5.4.1 Cement as a measure of construction

Cement emissions, though under 10 per cent of the carbon content of a windfall, provide the cleanest measure of the investment channel. Cement is heavy and inexpensive relative to its weight, so most cement consumed in a country is produced at domestic kilns, which makes cement emissions a measure of domestic construction. Moreover, its process emissions arise from the calcination of limestone and therefore depend on the amount of clinker produced, whatever the prices or mix of the fuels used to fire the kiln (Andrew 2019; Habert et al. 2020). Cement is also measured in three ways, through its emissions in the EDGAR database and in the Global Carbon Project emissions dataset (with a flag for the data source) and through production in physical tonnes, along with the capacity of every cement plant, so an accounting artefact in one measure would make the measures diverge (Table A14).

5.4.2 Cement, output and capacity

In EDGAR, cement process emissions show no response in the year of the shock but rise to 0.40 log points after two years and to 0.91 log points after five (Table 5; Figure 6). With the Global Carbon Project series the responses are 0.56 and 1.13, and with cement production in tonnes they are 0.48 and 1.11. Since cement emissions per unit of real GDP rise by 0.35 log points after five years, windfalls stimulate construction more than they stimulate the economy. For all construction materials together, process emissions rise by 0.73 log points after five years, and the pre-shock coefficients are small and insignificant for every series. A typical windfall raises cement emissions after five years by about 8 per cent.

Despite this increase in output, clinker capacity commissioned in the five years following the windfall does not increase (Table 5, Panel B). Among plants commissioned since 2005,

Table 5: Cement emissions, construction materials and new kiln capacity

	Cement CO_2 (EDGAR)	Cement CO_2 (GCP)	Cement output	Cement CO_2 / GDP	Construction materials
Panel A. Local projections					
Pre-period, $h = -2$	0.153	0.009	0.035	0.201	−0.014
	(0.127)	(0.143)	(0.110)	(0.132)	(0.063)
$h = 0$	0.127	0.098	0.080	−0.003	0.190
	(0.107)	(0.111)	(0.097)	(0.108)	(0.215)
$h = 2$	0.397**	0.563***	0.477***	0.123	0.223
	(0.151)	(0.183)	(0.132)	(0.137)	(0.203)
$h = 3$	0.528***	0.726***	0.642***	0.195	0.456*
	(0.148)	(0.200)	(0.139)	(0.134)	(0.232)
$h = 5$	0.906***	1.125***	1.113***	0.349**	0.733**
	(0.156)	(0.248)	(0.238)	(0.129)	(0.356)
$h = 8$	0.809***	0.990***	1.229***	0.291	0.618
	(0.281)	(0.265)	(0.265)	(0.229)	(0.453)
Observations ($h = 5$)	4,357	3,079	4,239	4,320	4,743
Panel B. New clinker capacity	*Plants since 2005*	*Plants since 1990*			
Windfall shock	−1.769**	−0.895			
	(0.773)	(1.231)			
Observations	881	2,134			

Notes: Panel A reports local projections for cement process CO_2 from EDGAR and from the Global Carbon Project (interpolated years removed), cement production in tonnes, cement process CO_2 per unit of real GDP, and process CO_2 from all construction materials. Panel A follows the sample and estimation of Table 2, and both panels use its significance levels. Panel B reports Poisson regressions of clinker capacity commissioned in $t + 1$ to $t + 5$ on the windfall shock in t (Global Energy Monitor cement tracker), for plants that started operating since 2005 and since 1990, with two shock lags, country and year fixed effects and errors clustered by country; the coefficient is a semi-elasticity.

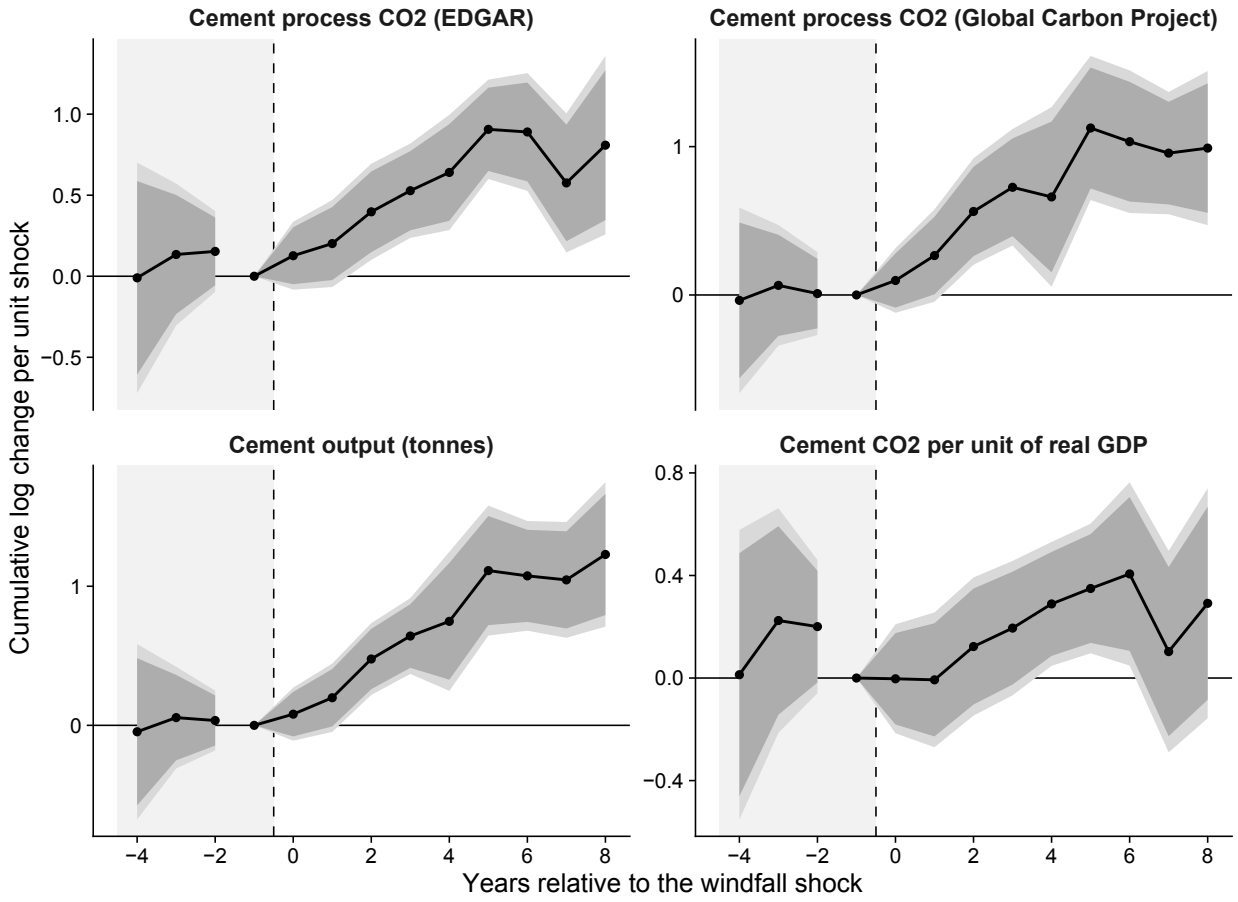


Figure 6: Cement emissions, cement output and cement emissions per unit of GDP after a fossil windfall. Bands and shading as in Figure 2

which are less likely to have closed and therefore less likely to be missing from the plant trackers, clinker capacity even declines: the Poisson semi-elasticity is −1.77 (standard error 0.77). Cement output increases by roughly as much as cement emissions while capacity does not increase, so the higher output comes from greater utilisation of existing capacity, as predicted by P3. Prices matter for heavy industries in the long run, when capacity is built, whereas windfalls change the utilisation of existing capital. This distinction also explains why short-run changes in prices, from fuel subsidy reforms or the 2022 spike in energy prices, have little effect on the output of existing plants (Table A16). Using satellite data on the building footprint, I find no evidence that settlements expand in response to windfalls (Table A15), though the building layer does not cover roads and other infrastructure and is interpolated between years of observation. Windfalls raise emissions by financing public works projects and filling installed kiln capacity, while booms do not increase capacity and busts do not lead to kiln closures.

5.5 Threats to identification

The responses of total CO_2 and cement emissions withstand the main identification concerns (Table 6). After exposure is interacted with world GDP growth, changes in the US real interest rate, China's cement cycle, a set of metals price innovations and an exposure-specific linear trend, the five-year response is 0.27 for total CO_2 and 0.99 for cement, while adding region-by-year fixed effects, which restricts comparisons to countries in the same region and year, brings them to 0.24 and 0.72. Accounting for battle deaths, dropping sanctioned economies and Kuwait or measuring exposure from 15 to 11 years before the shock produces coefficients of a similar magnitude or higher, and dropping OPEC members and Russia, which may set prices, yields 0.20 and 1.21. A placebo exercise interacting the oil price innovations with metal rents in place of fossil rents produces negative coefficients, which indicates that the results do not reflect a generic response of resource-dependent economies to high oil prices. The results are similarly robust for emissions from road transport and buildings and for household consumption (Table A9).

With relatively few price surprises to identify the effects, I stress-test the source of variation by leaving out one year of surprises at a time: the five-year response of total CO_2 stays within 0.21 to 0.31 and is significant at the 5 per cent level in all cases, while the carbon content stays within 0.42 to 0.58 kg (Table A11). When entire episodes are left out, such as the 1986 oil price collapse, the Gulf war, the boom of 2003 to 2014 or the crises of 2020 to 2022, the five-year response of total CO_2 ranges from 0.17 to 0.32 and remains significant at the 5 per cent level, except without the oil price collapse of 2014 to 2016 ($p = 0.13$) or

Table 6: Identification checks for total and cement CO_2

Specification	Total CO_2, $h = 3$	Total CO_2, $h = 5$	Cement CO_2, $h = 3$	Cement CO_2, $h = 5$
Baseline	0.236^{**}	0.265^{***}	0.528^{***}	0.906^{***}
	(0.101)	(0.073)	(0.148)	(0.156)
Global-cycle controls	0.193^{***}	0.270^{*}	0.434^{**}	0.988^{***}
	(0.069)	(0.138)	(0.181)	(0.103)
Region × year FE and global-cycle controls	0.157^{**}	0.235^{***}	0.300^{*}	0.724^{***}
	(0.061)	(0.073)	(0.175)	(0.108)
Battle deaths control (1989 on)	0.279^{**}	0.248^{*}	0.682^{***}	1.043^{***}
	(0.112)	(0.130)	(0.213)	(0.309)
Excluding sanctioned economies and Kuwait	0.365^{***}	0.293^{***}	0.676^{***}	1.037^{***}
	(0.084)	(0.094)	(0.183)	(0.309)
Excluding OPEC and Russia	0.425^{***}	0.204^{***}	0.704^{*}	1.209^{*}
	(0.131)	(0.064)	(0.388)	(0.652)
Exposure measured $t-15$ to $t-11$	0.388^{***}	0.416^{***}	0.933^{***}	1.373^{***}
	(0.124)	(0.125)	(0.319)	(0.347)
Placebo: metal rents × oil innovation	-1.158	-1.219	-0.479	-1.160
	(0.689)	(1.037)	(0.366)	(0.705)
Supply news as instrument (outside OPEC)	0.135	-0.147	1.491^{*}	2.347^{*}
	(0.176)	(0.160)	(0.861)	(1.160)

Notes: Each cell is the coefficient on the windfall shock from a separate local projection of cumulative log total fossil CO_2 or cement process CO_2 (EDGAR). The baseline sample, estimation and significance levels are those of Table 2. Global-cycle controls interact the lagged rents share S with world GDP growth, the change in the US ten-year real rate, the log change in China's cement emissions, real-time innovations in metals prices and a linear trend; region × year effects use World Bank regions; battle deaths are from UCDP; the placebo replaces fossil rents with metal mineral rents; the supply-news row instruments the windfall shock with oil rents times time-aggregated Känzig (2021) shocks, countries outside OPEC.

without all years after 2014 (p = 0.07). Estimating the baseline regression at the level of the 105 fuel-year surprises, controlling for the level of exposure, with the standard errors of Borusyak et al. (2022), yields p-values of 0.018 for total CO_2 and less than 0.001 for cement (Table A12). Permuting the years of the price surprises gives a p-value of 0.004 for cement, 0.018 for road transport and 0.060 for total CO_2, while permuting exposure between countries in the same region gives p-values of 0.010 or less (Table A10). The most demanding specification includes exposure quintile by year fixed effects. Given the close correlation between the prices of the three fuels, only differences across fuels within a year are then left for identification. Under world prices, the coefficient on total CO_2 is 0.17 and insignificant, and the cement coefficient also falls (Table A12). Using the price benchmarks of the regional markets where countries sell restores some variation, and the response of road transport is then 0.63 (p = 0.01), but the estimates for total CO_2 (0.20, p = 0.12) and cement are still insignificant (Table A13). This specification is still the main open challenge for the research design.

5.6 The carbon ratchet

The shock is split into its positive and negative parts (Table 7; Figure 7), and, to facilitate comparison, the response to a negative shock is plotted multiplied by -1, so that the lines coincide when the response is symmetric. Three years after a positive windfall of one unit, emissions from cement, buildings and power rise by 1.67, 1.77 and 1.38, respectively, and household consumption by 0.74, all significant at the 5 per cent level or better. In contrast, a negative windfall of the same size has no significant effect on cement emissions, buildings emissions or household consumption, and power sector emissions, if anything, increase when the windfall is negative. I reject the null of equal responses at the 5 per cent level for cement, buildings and household consumption and at the 1 per cent level for power.

Public investment drops significantly in busts, but the size of the response is similar in booms, and I cannot reject the null of equality. Although governments cut investment in busts, the emissions that came with the boom do not decline. The intuition is that booms increase the stock of structures, vehicles and buildings, which generate emissions over a long period, whereas cutting investment in busts does not remove the existing stocks, as prediction P4 states. The increase in power emissions following negative windfalls may be driven by a separate mechanism specific to producer economies: when export prices fall, domestic oil and gas become cheaper to burn in power plants relative to exporting them. At the aggregate level, I find no significant asymmetry in total fossil CO_2, as sectors whose responses are symmetric or asymmetric in the opposite direction offset the ratchet. The carbon ratchet

Table 7: Responses to positive and negative windfall shocks ($h = 3$)

	Cement CO_2	Buildings CO_2	Power CO_2	Total CO_2	Household consumption	Public investment
Positive shock (boom)	1.671***	1.772**	1.378**	0.309	0.738***	2.188
	(0.532)	(0.715)	(0.586)	(0.367)	(0.165)	(1.617)
Negative shock (bust)	−0.158	−0.008	−0.647**	0.192	0.145	1.846**
	(0.419)	(0.270)	(0.309)	(0.218)	(0.144)	(0.893)
Equal responses, p	0.045	0.047	0.008	0.831	0.018	0.885
Observations	4,659	5,022	4,968	5,059	3,886	3,949

Notes: The windfall shock is split into its positive and negative parts in the baseline local projection at horizon three; the controls are the two lags of the full shock and two lags of the outcome change. Otherwise, the sample, estimation and significance levels are as in Table 2. A positive coefficient on the negative part means the outcome falls when the shock is negative; the p-value tests whether the two coefficients are equal.

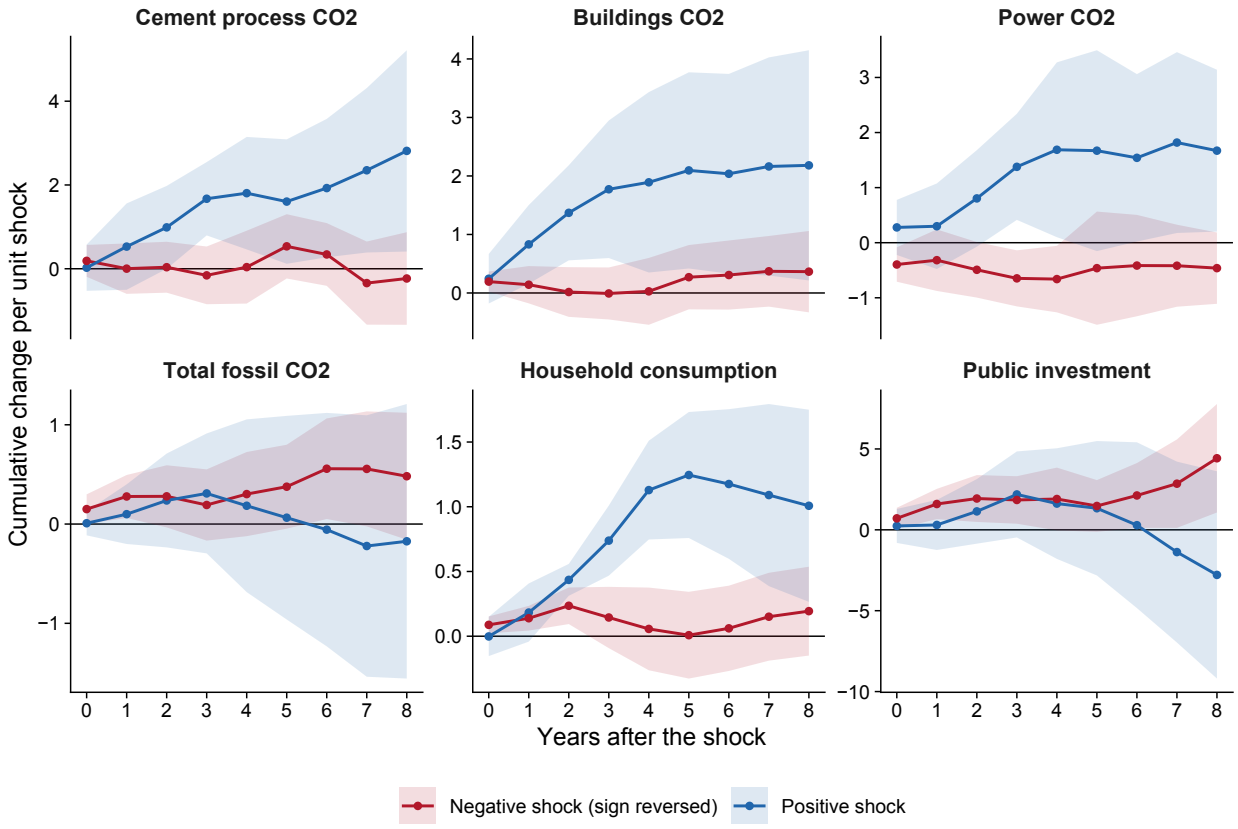


Figure 7: Responses to positive and negative windfall shocks. Negative responses carry their sign reversed; bands are 90 per cent confidence intervals

has important implications for the energy transition: as decarbonisation lowers fossil fuel revenues, public investment in producer economies would fall, but the emissions that past booms added in buildings, transport and power would remain, and the price spikes that occur during the transition could raise them further.

5.7 Institutions

For the comparison of institutions, I interact the shock simultaneously with pre-shock indicators for a fiscal rule in force, autocracy, and fuel underpricing, including all lower-order terms (Table 8; Figure 8). I focus on the shocks of 1995 to 2016, when all three institutions are observed, which covers between 81 and 90 countries, depending on the outcome. I centre each moderator at its mean in the estimation sample, so that the shock coefficient reflects the impact with average institutions (Balli and Sørensen 2013). With average institutions, a unit shock increases public investment by 7.5 log points after two years, and cement emissions by 2.0 log points after two years and by 2.5 after five. The public investment response after two years is 5.5, 4.9 and 4.1 log points smaller with a fiscal rule, autocracy and fuel underpricing, respectively, and the three interactions are jointly significant ($p = 0.01$). The interaction with fiscal rules survives after five years at the 10 per cent level, but the other two do not. Institutions do not modify the cement response. Autocracy decreases, and fuel underpricing increases, the total emissions response after two and five years respectively. The former finding is in line with Oyekola et al. (2024), who show that commodity booms raise emissions in democracies only. The latter interaction is insignificant when standard errors are clustered by country only ($p = 0.11$).

Fiscal rules, by forcing governments to save part of the windfall, and fuel subsidies, by funnelling rents away from investment budgets before they can be spent, move absorption out of public investment, as predicted by P5. However, there is no evidence of a smaller construction response or of a lower overall carbon content of windfalls, and where fuel is underpriced, the response of total emissions is, if anything, greater. These estimates are conditional associations, sensitive to how the sample is constructed and to the covariance estimator.

6 The climate cost of windfalls

I multiply the cumulative response estimates (horizons 0 to 8) by the windfall shock history for each country and sum the implied emissions, under the assumption that the effects are zero after eight years. Windfalls increased emissions by 2.7 Gt of fossil carbon dioxide over

Table 8: Fiscal rules, autocracy, fuel underpricing and the windfall response

	Public inv., $h = 2$	Public inv., $h = 5$	Cement CO_2, $h = 2$	Cement CO_2, $h = 5$	Total CO_2, $h = 2$	Total CO_2, $h = 5$
Windfall shock, average institutions	7.463***	1.727	2.035*	2.549***	0.503*	0.285
	(1.695)	(3.433)	(1.148)	(0.811)	(0.273)	(0.325)
Windfall × fiscal rule	−5.493***	−7.093*	−2.008	−0.831	−0.288	−0.418
	(1.841)	(3.821)	(1.644)	(0.995)	(0.405)	(0.666)
Windfall × autocracy	−4.883**	−0.982	−0.811	0.123	−0.491***	−0.588
	(1.833)	(2.864)	(1.277)	(1.045)	(0.127)	(0.366)
Windfall × underpriced fuel	−4.069***	−3.503	−0.515	−1.019	0.171	0.786**
	(1.376)	(2.387)	(1.203)	(1.182)	(0.213)	(0.378)
Joint test of interactions, p	0.010	0.354	0.427	0.812	0.000	0.114
Country-clustered p, × fiscal rule	0.004	0.033	0.161	0.460	0.322	0.519
Country-clustered p, × autocracy	0.013	0.740	0.494	0.899	0.045	0.055
Country-clustered p, × underpriced fuel	0.018	0.243	0.687	0.404	0.541	0.109
Countries	88	87	81	81	90	90
Observations	1,545	1,367	1,322	1,302	1,576	1,573

Notes: The windfall shock is interacted at the same time with three moderators measured before the shock, each centred at its mean in the estimation sample, so the first row is the response at average institutions and the interaction coefficients do not depend on the centring (Balli and Sørensen 2013). Where the two-way clustered variance matrix is not positive semi-definite, it is repaired with the eigenvalue adjustment of Cameron et al. (2011), applied in the centred parametrisation; because that repair is not invariant to reparametrisation, it can change the interaction standard errors, so the table also reports p-values with clustering by country alone, whose variance matrix needs no repair. The fiscal rule indicator equals one if a national or supranational rule was in force at $t-1$ (IMF Fiscal Rules Dataset 1985 to 2024); autocracy means Polity2 at or below zero at $t-1$; underpriced fuel means a gasoline net tax below zero on average over $t-5$ to $t-1$. Every lower-order term is included, among them the moderator main effects and the moderator × shock-lag terms; shocks from 1995 to 2016, the years in which all three moderators are observed; cement CO_2 from the Global Carbon Project without interpolated years. The sample, estimation and significance levels otherwise follow Table 2. The joint test is a Wald test of the three interaction coefficients.

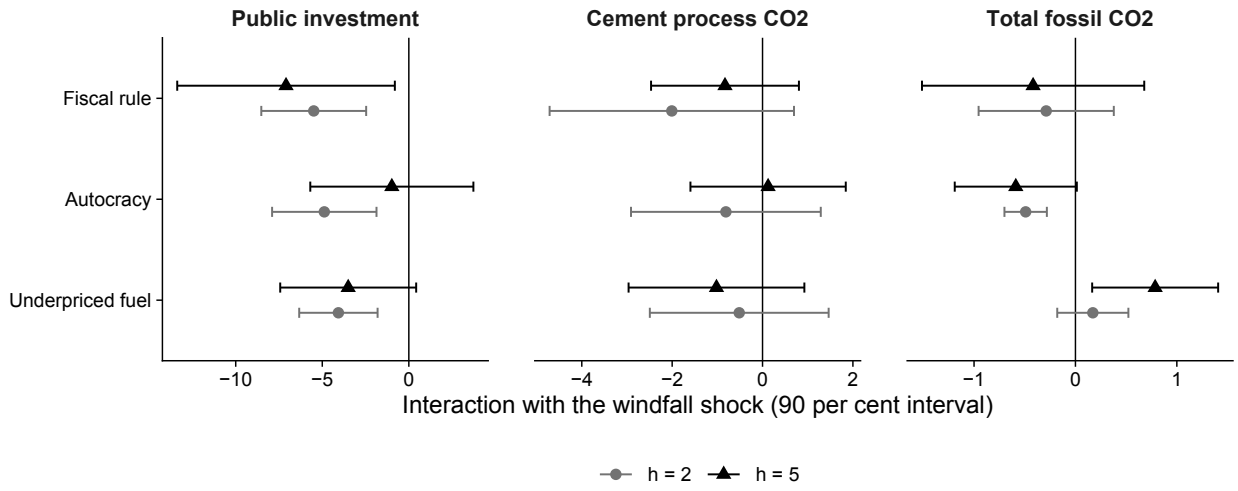


Figure 8: Interactions of the windfall shock with fiscal rules, autocracy and fuel underpricing. Whiskers: 90 per cent confidence intervals

the boom of 2003 to 2014, or 0.7 per cent of the sample's total fossil emissions (Table 9; Figure 9). The 90 per cent confidence interval is 1.3 to 4.1 Gt. For the 32 economies with rents of at least 10 per cent of GDP in some year during the boom, this rise was 3.5 per cent of emissions (Table 9). This increase includes 1.0 Gt in the road transport sector, 0.7 Gt in buildings, and 0.21 Gt in cement process emissions, or 0.9 Gt for cement if I use the separate responses for boom and bust from Table 7 (Table 9). The gap arises because the boom consisted of repeated positive windfalls, which were not subsequently reversed by negative windfalls. Among producers, the largest contributors were Saudi Arabia, Russia, Iran and Iraq.

China also made a large contribution (0.4 Gt, or 0.4 per cent of its emissions; Figure 9b) due to its large emissions and coal rents that were sizeable relative to its GDP; without China, the implied increase is 2.3 Gt. The mean social cost of carbon estimated by Rennert et al. (2022) is $185 per tonne at 2020 prices. If I apply this to the increase in emissions during the boom, the implied climate damage amounts to about $500 billion, with the 5th and 95th percentiles of the distribution implying $118 billion and $1.1 trillion of damage. Other recent estimates of the macroeconomic damage from temperature shocks are even larger (Bilal and Känzig 2024). At the mean social cost of carbon, each dollar of windfall revenue thus generates about 10 cents in climate damage (Table 9), which is not recorded in the producing economies' public accounts.

Table 9: Emissions attributable to windfalls and their climate damage, 2003 to 2014

Emissions	Attributable CO_2 (Mt)	90% interval (Mt)	Share of emissions (%)	Damage at USD 185/t (bn)	Damage at USD 44 to 413/t (bn)
Total fossil CO_2	2,676	[1,297, 4,101]	0.7	495	118 to 1,105
of which road transport	995	[496, 1,466]	1.6	184	44 to 411
of which buildings	682	[283, 1,062]	1.8	126	30 to 282
of which construction materials (process)	238	[−19, 469]	1.1	44	10 to 98
Cement process CO_2	211	[106, 315]	1.5	39	9 to 87
Cement process CO_2, boom and bust responses	897	n.a.	6.6	166	39 to 371

Notes: Attributable emissions apply the estimated cumulative log responses at horizons 0 to 8 to each country's history of windfall shocks, assuming no effect beyond eight years, and sum the implied emissions over 2003 to 2014; sector rows use their own responses, so they need not add up to the total. The last row uses separate responses to positive and negative shocks (Table 7). Intervals take 1,000 draws in which every horizon's coefficient moves by the same number of standard errors, which is conservative relative to independent draws. In the 32 economies with rents of at least 10 per cent of GDP in some year of the boom, attributable emissions are 3.5 per cent of their fossil CO_2; without China, total attributable emissions are 2,290 Mt. The sample is that of Table 2. Damages, in billions of 2020 US dollars, use the social cost of carbon of Rennert et al. (2022), USD 185 per tonne at the mean and USD 44 and 413 at the 5th and 95th percentiles. At the mean, the carbon content of Table 4 (0.52 kg per dollar) implies climate damage of about 10 US cents per dollar of windfall revenue. No significance tests are reported.

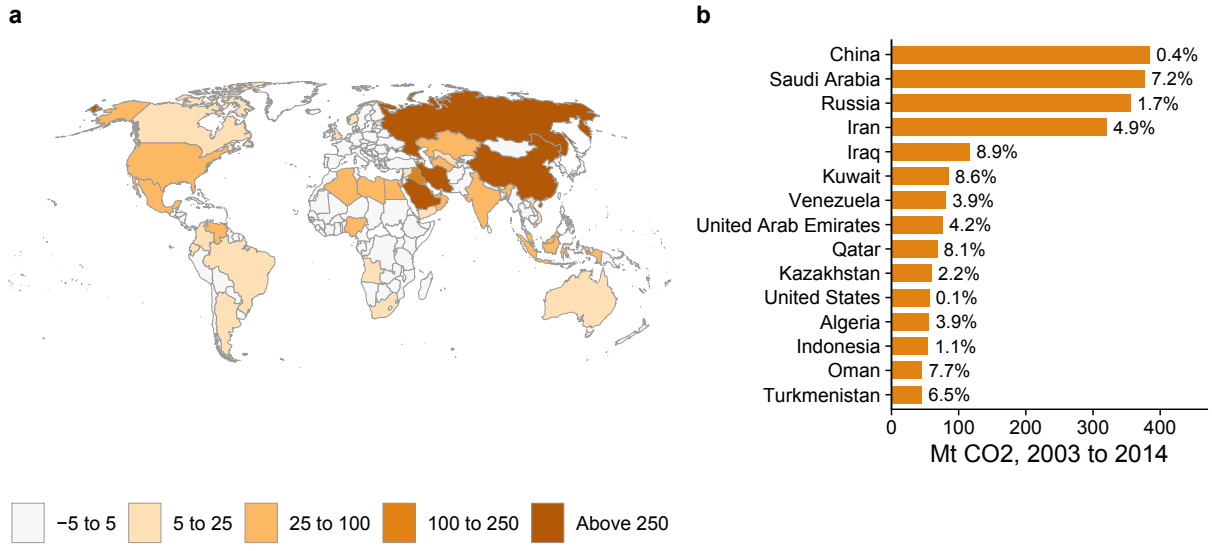


Figure 9: Fossil CO_2 attributable to windfalls by country (a) and the fifteen largest contributors (b), 2003 to 2014. Labels in (b): attributable emissions as a share of the country's fossil CO_2

7 Discussion

7.1 Transition risk and producer economies

Several papers have argued that the ongoing energy transition will lead to lower and more volatile fossil revenues, which can be thought of as a source of transition risk for the asset values and public finances of fossil producers (Battiston et al. 2017; Mercure et al. 2018; van der Ploeg and Rezai 2020; Peszko et al. 2020). My results suggest that, in addition to average revenues, the emission intensity of revenue spikes matters for emissions in producer economies. Producer economies tend to spend their revenue spikes on carbon-intensive consumption and construction (Sections 5.2 and 5.4). Moreover, the emission impact of spending revenue spikes on durables does not reverse in downturns. As a result, economies that experience revenue booms and busts during the transition end up with lower revenues and, in the building, power, and construction sectors, emit more than would be implied by their average revenue levels. My findings also highlight a channel through which revenue volatility induces emission lock-in without new power plants or kilns, which is not captured by engineering studies on committed emissions (e.g. Tong et al. 2019), namely through the use of durable stocks such as vehicles, buildings, and existing industrial capacity. My results also qualify the climate effects of high fossil fuel prices. While an exogenous supply shock that increases fossil fuel prices leads to reduced fuel use in importing countries (Känzig 2023; Metcalf and Stock 2023), the corresponding increase in revenues in exporting countries leads in part to higher emissions domestically. This form of income leakage has received little attention in prior work on carbon leakage, which has primarily focused on the relocation of production.

7.2 Policy

My results imply that the management of fossil revenues is an important part of climate policy in producer economies. While fiscal rules that force governments to save revenue windfalls are associated with a smaller public investment response to a boom, several recent rules exempt public investment from their limits (Alonso et al. 2025), keeping that channel open. My results do not suggest that fiscal rules reduce the impact of windfalls on construction activity or the total carbon content of windfalls, most of which goes to consumption, so they are not a substitute for policies that reduce the carbon content of spending. Policies that reduce the emission intensity of durables, such as building codes, vehicle efficiency standards, and public transport investment instead of road construction, can limit the ratchet effect, because it operates through durable stocks. They are also cheaper to implement during revenue booms than to change afterwards. At the same time, fuel subsidy reforms and

windfall revenue management policies interact: a subsidy reform that releases rents with no rules on how they are used may move emissions from fuel use to construction.

7.3 Discoveries, disasters, plant data and limits

Giant oil and gas discoveries have no impact on construction activity or emissions in this sample (Table A19), consistent with governments spending rents as they come in. Large natural disasters in the EM-DAT climate disaster database (Delforge et al. 2025) are associated with higher cement emissions in high fossil-rent economies, although only three of them hit these economies (Table A20). A monthly satellite measure of activity at cement and steel plants tracks official production statistics at the annual level, but the apparent responses of this measure to windfalls disappear when I control for the update to the satellite retrieval processor in July 2022 (v2.4 of the TROPOMI nitrogen dioxide (NO_2) product; Copernicus Sentinel-5P 2021), so I use the plant data as a contribution to measurement (Tables A17 and A18; Figures A2 and A3).

The empirical design estimates responses to unexpected price movements (weighted by predetermined exposure), which are not the same as the effect of permanent rent changes. Under supply news, the responses of macroeconomic aggregates, investment, cement, buildings and road transport are of the same size as in the main design, as is the EIA energy CO_2 response averaged over years 0 to 5, but the EDGAR response of total emissions is significant only in the first two years or with four lags, partly because power does not respond (Section 5.3). These results are consistent with a channel where windfalls raise emissions through absorption, and the carbon content requires an exclusion restriction, which is supported by the checks in Section 4.2. Nevertheless, there are caveats: the variation is time-series (only a few dozen years, with correlated fuel price surprises), so there is little variation left in the specification with the most demanding fixed effects; emissions data are at the country level; and the results on institutions are merely associative. The calculation of the climate cost assumes that the response is linear and zero beyond eight years.

8 Conclusion

Fossil revenue windfalls flow through governments' budgets and, in part, into carbon emissions. Most of this revenue is saved abroad in the first year, but within a couple of years it is largely spent on consumption, investment and imports, and emissions rise in road transport, buildings, power generation and industry. Each dollar of windfall revenue is associated with about 0.5 kg of additional domestic carbon dioxide emissions in two distinct datasets

on national carbon emissions. Changes in cement, measured in three ways, are consistent with an investment channel that runs through existing cement kilns. Booms raise carbon emissions by raising the stocks of carbon-emitting durables, and this increase remains even when revenues fall. Fiscal rules and fuel pricing policies are associated with a lower public investment response to fossil fuel revenues, but they are not associated with a lower emissions response. For economies that produce fossil fuels, how they manage this source of revenue is an important part of their climate policy. Moreover, the expected volatility of these revenues as the world transitions away from carbon emissions will raise emissions in these countries unless the spending is appropriately managed.

Data and code availability. All data used in this study are publicly available. A replication package, including computer code (a master file that generates all tables and figures), raw data (where the licence permits) and derived satellite data panels, will be made available in a public repository with a persistent identifier on publication. Two of the datasets, EM-DAT and the UNU-WIDER Government Revenue Dataset, require free registration with the data providers to download. The README file in the replication package includes information on the source, licence and access for each file.

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Appendix

A Data

All sources are public (Table A1). The replication package includes every raw file whose terms allow redistribution, among them the Global Energy Monitor trackers (CC BY 4.0) and the IMF datasets, whose terms allow distribution with attribution. The IEA-EDGAR CO_2 data (CC BY-NC-ND 4.0) and the discoveries data of Arezki et al. (2017) are downloaded by script, and EM-DAT and the UNU-WIDER Government Revenue Dataset are obtained after free registration with their providers. The package supplies the national built-up totals computed from the Global Human Settlement Layer, because the 1 km grids, open data of the European Commission Joint Research Centre, take 2.7 GB; an optional step downloads them.

Table A1: Data sources and definitions

Variable	Source	Coverage
Fossil CO_2 by sector (IPCC 2006 categories) and total; methane from fuel extraction	EDGAR 2026 (Crippa et al. 2026)	Country-year, 1970 to 2025
Cement CO_2 and production, with source flags	Global Carbon Project cement dataset, release 260526 (Andrew 2019)	Country-year, to 2025
Oil, gas, coal and mineral rents; GDP in current and constant dollars	World Development Indicators (World Bank 2026a)	Country-year, 1970 to 2024
Household and government consumption, fixed investment, exports, imports, current account	World Development Indicators (World Bank 2026a; constant 2015 dollars; current account in per cent of GDP)	Country-year, 1970 to 2024
Public and private investment	IMF Investment and Capital Stock Dataset 2021 (International Monetary Fund 2021)	Country-year, 1960 to 2019
Government revenue: total, resource and non-resource	UNU-WIDER Government Revenue Dataset 2025 (UNU-WIDER 2025), merged general or central government series	Country-year, 1980 to 2023
Energy CO_2 and consumption of oil products, gas, coal and electricity	US Energy Information Administration international energy statistics (U.S. Energy Information Administration 2026)	Country-year, 1980 to 2024
World fossil and metals prices	World Bank Commodity Price Data, the Pink Sheet (World Bank 2026b), deflated by the US consumer price index (Federal Reserve Bank of St. Louis 2026)	Monthly, 1960 to 2026
US real interest rate	10-year Treasury yield (DGS10) less US consumer price inflation (Federal Reserve Bank of St. Louis 2026)	Daily, 1962 to 2026
Oil supply news shocks	Känzig (2021), vintage 2025M12	Monthly, 1975 to 2025

Continued on the next page

Table A1 (continued)

Variable	Source	Coverage
Built-up surface (building footprint), total and non-residential	Global Human Settlement Layer GHS-BUILT-S R2023A, 1 km grids (Pesaresi 2023; Pesaresi et al. 2024), summed within Natural Earth country borders (Natural Earth 2026)	Five-year epochs, 1975 to 2020
Cement plants: capacity, start year, coordinates	Global Cement and Concrete Tracker, July 2026 V1 (Global Energy Monitor 2026a)	Plant level
Steel plants: capacity and start year	Global Iron and Steel Tracker, September 2026 V1 (Global Energy Monitor 2026b)	Unit level
Giant oil and gas discoveries	Arezki et al. (2017)	Country-year, 1970 to 2012
Fiscal rules	IMF Fiscal Rules Dataset 1985 to 2024 (Alonso et al. 2025)	Country-year
Political regime	Polity2 score of Polity5 (Marshall and Gurr 2020), from the replication data of Martinez-Alvarez et al. (2022)	Country-year
Gasoline net tax	Ross et al. (2017), from the replication data of Martinez-Alvarez et al. (2022), extended with the replication data of Mahdavi et al. (2025) and the Global Fuel Prices Database (World Bank 2026c)	Country-year, 1990 to 2023
Armed conflict	UCDP Organized Violence country-year 26.1 (Davies et al. 2026)	Country-year, 1989 to 2025
Disasters	EM-DAT (Delforge et al. 2025)	Event level, 2000 to 2026
Plant activity	Sentinel-5P TROPOMI tropospheric NO_2 (Copernicus Sentinel-5P 2021) via Google Earth Engine (Gorelick et al. 2017)	Plant-month, 2018 to 2026
Surface albedo at plant sites	MODIS MCD43A3 V061 white-sky shortwave albedo, 2019 mean (Schaaf and Wang 2021)	500 m grid
Plant locations for the satellite panel	Spatial Finance Initiative cement and iron and steel asset databases (McCarten et al. 2021; Tkachenko et al. 2023)	Plant level
COVID-19 stringency	Oxford COVID-19 Government Response Tracker, national stringency index (Hale et al. 2021)	Country-day, 2020 to 2022

Notes: The table lists each variable, its source and coverage; full references are in the reference list of the paper, and the README of the replication package gives the licence and access of every file. The IEA-EDGAR CO_2 data (CC BY-NC-ND 4.0), EM-DAT, the UNU-WIDER Government Revenue Dataset and the discoveries data of Arezki et al. (2017) are not redistributed and are obtained from their providers, and the table reports no estimates.

B Transmission and the government budget

Private investment from the IMF Investment and Capital Stock Dataset rises after a windfall, but its pre-period coefficient two years before the shock is negative and significant, so the

main text uses total fixed investment from the World Development Indicators, which has flat pre-periods (Table A2). Imports rise strongly and exports do not move. Refining emissions rise in the year of the shock, while chemicals emissions do not respond significantly. Methane from fuel extraction does not respond.

Table A2: Private investment, trade, refining, chemicals and methane

Horizon	Private investment	Imports	Exports	Refining CO_2	Chemicals CO_2	Methane, fuel extraction
Pre-period, $h = -3$	-0.362^{**}	-0.077	-0.227	-0.109	-0.186^{**}	-0.042
	(0.133)	(0.167)	(0.135)	(0.181)	(0.072)	(0.162)
Pre-period, $h = -2$	-0.294^{***}	-0.147^{*}	-0.151	-0.043	0.003	-0.081
	(0.047)	(0.083)	(0.104)	(0.125)	(0.035)	(0.111)
$h = 0$	0.041	0.358^{***}	0.064	0.232^{**}	0.028	0.151
	(0.170)	(0.090)	(0.149)	(0.110)	(0.142)	(0.105)
$h = 2$	1.037^{***}	1.027^{***}	0.156	-0.031	0.020	-0.152
	(0.174)	(0.179)	(0.163)	(0.167)	(0.246)	(0.126)
$h = 3$	0.974^{***}	0.874^{***}	-0.051	0.133	0.233	-0.234
	(0.249)	(0.245)	(0.205)	(0.151)	(0.300)	(0.159)
$h = 5$	0.700^{*}	0.993^{***}	-0.166	0.239	0.552	-0.111
	(0.412)	(0.292)	(0.197)	(0.166)	(0.403)	(0.298)
$h = 8$	0.865	1.092^{***}	-0.172	-0.055	0.340	-0.165
	(0.569)	(0.317)	(0.228)	(0.211)	(0.403)	(0.314)
Observations ($h = 5$)	3,648	3,734	3,734	3,462	4,743	4,639

Notes: Each coefficient is the cumulative log change in the outcome from $t-1$ to $t+h$ per unit windfall shock, from a separate local projection. Private investment is from the IMF Investment and Capital Stock Dataset and shows a pre-period coefficient at $h = -2$, so it is not used in the main text; imports and exports are World Development Indicators volumes; refining is IPCC 2006 category 1.A.1.b and 1.A.1.c, chemicals 2.B and 2.D, and methane from fuel extraction 1.B (EDGAR 2026). The sample is country-years with total fossil CO_2 of at least 1 Mt in $t-1$, shocks from 1981 to 2024. Estimation is by OLS local projections with country and year fixed effects and two lags of the shock and of the outcome change; standard errors in parentheses are two-way clustered by country and year. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Government resource revenue tracks fossil rents closely, total revenue follows it, and non-resource revenue barely moves (Table A3). The revenue data come from the UNU-WIDER Government Revenue Dataset 2025 (UNU-WIDER 2025), which separates resource revenue from other government revenue. Revenue items are converted to local currency with the dataset's own GDP series and expressed relative to GDP before the shock, the same scale as the fossil-rent response, so a coefficient of one means an amount equal to the whole windfall. The capture share, estimated by regressing cumulative resource revenue on cumulative rents with the windfall shock as the instrument, is close to one after five years, and excluding country-years that the dataset flags for questionable accuracy leaves the estimates unchanged.

Table A3: Government revenue after a fossil windfall

Horizon	Fossil rents (WDI)	Government resource revenue	Government non-resource revenue	Total government revenue
Panel A. Windfall shock (price innovations)				
Pre-period, $h = -2$	0.019	0.012	-0.014^{**}	0.016
	(0.027)	(0.016)	(0.006)	(0.015)
$h = 0$	0.797^{***}	0.564^{***}	0.007	0.592^{***}
	(0.117)	(0.088)	(0.014)	(0.096)
$h = 1$	0.742^{***}	0.769^{***}	0.073	0.849^{***}
	(0.218)	(0.113)	(0.047)	(0.113)
$h = 2$	0.691^{***}	0.735^{***}	0.090	0.860^{***}
	(0.252)	(0.150)	(0.064)	(0.171)
$h = 3$	0.757^{***}	0.798^{***}	0.173^{*}	0.835^{***}
	(0.249)	(0.177)	(0.097)	(0.200)
$h = 5$	0.599^{*}	0.969^{***}	0.227	0.903^{***}
	(0.314)	(0.245)	(0.165)	(0.229)
Observations ($h = 3$)	1,447	1,305	1,248	1,315
Panel B. Windfall shock instrumented by oil supply news, outside OPEC				
Pre-period, $h = -2$	0.063^{*}	0.030	0.015	0.038
	(0.037)	(0.025)	(0.010)	(0.031)
$h = 0$	0.455	0.579^{***}	0.068	0.657^{***}
	(0.321)	(0.166)	(0.068)	(0.191)
$h = 1$	-0.235	0.104	0.004	0.150
	(0.547)	(0.424)	(0.152)	(0.355)
$h = 2$	-0.511	0.292	-0.230	-0.015
	(0.674)	(0.235)	(0.176)	(0.410)
$h = 3$	-0.074	0.215	-0.169	-0.271
	(0.353)	(0.238)	(0.220)	(0.423)
$h = 5$	-0.443	-0.199	-0.495	-0.378
	(0.857)	(0.572)	(0.431)	(0.608)
Observations ($h = 3$)	1,010	868	815	896

Notes: Each coefficient is the change in the revenue item from $t-1$ to $t+h$, in local currency and divided by GDP in $t-1$, per unit windfall shock, so a coefficient of one means an amount equal to the whole windfall. Government revenue is from the UNU-WIDER Government Revenue Dataset 2025 (merged series, general government where available); fossil rents are World Development Indicators oil, gas and coal rents converted to local currency with the same GDP series. All columns use the country-years covered by the resource revenue series. Panel A is estimated by OLS, Panel B by two-stage least squares with oil rents times time-aggregated Känzig (2021) supply news as the instrument, countries outside OPEC. Controls are two lags of the shock and the lagged change of the outcome, with country and year fixed effects; country-years whose change exceeds GDP are dropped. Standard errors in parentheses are two-way clustered by country and year. Regressing cumulative government resource revenue on cumulative fossil rents, with the windfall shock as the instrument, gives a capture share of 0.761 over horizons 0 to 3 (standard error 0.098, first-stage F 27.3, Anderson-Rubin $p < 0.001$, 62 countries) and 1.007 over horizons 0 to 5 (standard error 0.094, first-stage F 23.0, Anderson-Rubin $p < 0.001$, 59 countries). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C Carbon content and independent energy data

The carbon content of windfall revenue lies between 0.48 and 0.59 kg of CO_2 per dollar across horizons, trimming rules, global-cycle controls and the exclusion of China, and excluding OPEC members and Russia raises the point estimate to 0.81 but leaves it imprecise (Table A4).

Table A4: Carbon content of windfall revenue across horizons, trims and samples

Specification	Total fossil CO_2	Road transport	Buildings	Power	Construction materials	Cement
Baseline (horizons 0 to 8, 1 per cent trim)	0.522**	0.139***	0.054**	0.117	0.036***	0.031***
First-stage F 11.5	(0.178)	(0.035)	(0.021)	(0.060)	(0.013)	(0.011)
Horizons 0 to 5	0.484***	0.135***	0.050*	0.083	0.027***	0.029***
First-stage F 24.8	(0.154)	(0.034)	(0.025)	(0.065)	(0.011)	(0.011)
Trim 0.5 per cent	0.502***	0.131***	0.045*	0.081	0.028***	0.027***
First-stage F 11.6	(0.154)	(0.033)	(0.020)	(0.059)	(0.011)	(0.010)
Trim 2 per cent	0.574*	0.162***	0.055	0.121	0.040***	0.037***
First-stage F 9.8	(0.223)	(0.039)	(0.030)	(0.090)	(0.015)	(0.012)
Global-cycle controls	0.593**	0.163***	0.067**	0.141	0.048***	0.041***
First-stage F 9.9	(0.224)	(0.042)	(0.033)	(0.071)	(0.014)	(0.010)
Excluding OPEC and Russia	0.813	0.162	0.104	0.106	0.040	0.044**
First-stage F 5.7	(0.703)	(0.088)	(0.150)	(0.209)	(0.026)	(0.027)
Excluding China	0.513***	0.140***	0.051*	0.097	0.034***	0.029***
First-stage F 11.6	(0.159)	(0.035)	(0.025)	(0.052)	(0.013)	(0.010)

Notes: Each cell is the instrumental-variables coefficient of cumulative extra CO_2 of the sector (tonnes) on cumulative extra fossil rent revenue (US dollars), both relative to $t-1$ and divided by GDP in $t-1$, with the windfall shock as instrument, expressed in kg CO_2 per dollar; controls are two lags of the shock and lagged changes in emissions and rents, with country and year fixed effects. The sample is country-years with total fossil CO_2 of at least 1 Mt in $t-1$, shocks from 1981 to 2024. Standard errors in parentheses are two-way clustered by country and year; stars are from Anderson-Rubin tests (the reduced-form test in a just-identified model); the first-stage F is the clustered Wald statistic for the total CO_2 regression of each row. * $p<0.10$, ** $p<0.05$, *** $p<0.01$.

Weak-instrument-robust inference gives a 95 per cent Anderson-Rubin set from 0.20 to 0.89 kg per dollar for the baseline (Table A5). Each set is obtained by inverting the reduced-form test for every candidate carbon content with the same two-way clustered variance as the main estimates; robustness to a weak instrument does not remove the assumptions behind that clustered approximation. In economies that underpriced fuel before the shock, where world price surprises do not reach domestic fuel prices, the estimate is 0.36 kg, and in economies with taxed fuel it is 0.27 kg, with a set from -0.32 to 0.78 that includes zero. Using one fuel's price surprises at a time as the instrument gives 0.50 kg for oil and 0.74 kg

for gas, and a test of equal carbon content across the three fuels does not reject at the five per cent level (p = 0.08). Excluding OPEC members and Russia gives a wide, bounded set that includes zero, and with supply news as the instrument the first stage for cumulative revenue is too weak to estimate the carbon content. A placebo that interacts the oil price surprise with metal rents moves neither cumulative fossil revenue nor cumulative emissions (reduced-form coefficients −2.11, standard error 2.75, and 0.01, standard error 6.33).

The emissions results also hold in the international energy statistics of the US Energy Information Administration (EIA), which are built from measured consumption of each fuel and product and in which the EIA computes energy CO_2 itself (Table A6). EIA energy CO_2 and EDGAR fossil CO_2 net of process and fugitive emissions correlate at 0.99 in log levels and 0.39 in annual log changes. Energy CO_2 and measured consumption of oil products, gasoline and diesel, and electricity rise after windfalls, and with supply news as the instrument energy CO_2 rises significantly in each of the first three years.

D Supply-driven windfalls

With four lags of the windfall shock and of the outcome, the supply-news response of total CO_2 is significant after two years at the ten per cent level and after three years at five per cent, and the pre-period coefficients for total CO_2, road transport and buildings are small, while those for cement, household consumption and public investment remain negative (Table A7, Panel A). The simple annual sum of the monthly news shocks, which ignores the month in which a shock arrives, gives the same pattern as a reduced form, with coefficients on the scale of that variable (Panel B).

Averaging the supply-news responses over horizons pools the horizons and raises power for slow-moving aggregates (Table A8).

E Identification and inference

The identification checks of Table 6 also hold for road transport, buildings and household consumption (Table A9).

Randomisation inference re-estimates the five-year responses after permuting the years of the fuel price innovations (500 draws) and after permuting exposure histories across countries within World Bank regions (300 draws). It rejects the null of no effect for cement, road transport and, at the ten per cent level, total CO_2 when the years of the price surprises are permuted, and for all three when exposure histories are permuted across countries within regions (Table A10, Figure A1).

Table A5: Carbon content of windfall revenue: weak-instrument-robust inference and checks on the exclusion restriction

Specification	kg CO_2 per USD	Wald 95% interval	Anderson-Rubin 95% set	First-stage F	Observations
Baseline (horizons 0 to 8)	0.522**	[0.17, 0.87]	[0.20, 0.89]	11.5	4,215
	(0.178)				
Horizons 0 to 5	0.484***	[0.18, 0.79]	[0.17, 0.83]	24.8	4,679
	(0.154)				
Global-cycle controls	0.593**	[0.15, 1.04]	[0.14, 1.21]	9.9	4,215
	(0.226)				
Excluding OPEC and Russia	0.813	[−0.57, 2.19]	[−3.73, 2.10]	5.7	3,690
	(0.703)				
Excluding China	0.513***	[0.20, 0.82]	[0.22, 0.90]	11.6	4,181
	(0.159)				
Fuel taxed before the shock (1995 on)	0.266	[−0.19, 0.72]	[−0.32, 0.78]	11.1	2,052
	(0.231)				
Fuel underpriced before the shock (1995 on)	0.362**	[0.12, 0.60]	[0.08, 0.78]	9.0	334
	(0.124)				
Supply news as instrument, outside OPEC (horizons 0 to 5)	−1.132	[−4.66, 2.39]	$(-\infty, 0.88] \cup [6.64, \infty)$	2.1	4,128
	(1.799)				
Oil price surprises only as instrument	0.498**	[0.13, 0.86]	[0.16, 0.94]	10.6	4,215
	(0.187)				
Gas price surprises only as instrument	0.736*	[0.21, 1.26]	[−0.18, 1.18]	11.7	4,215
	(0.267)				
Coal price surprises only as instrument	4.005	[0.50, 7.51]	[−8.81, 8.74]	4.8	4,215
	(1.790)				
Construction materials (process), baseline	0.036***	[0.01, 0.06]	[0.02, 0.09]	10.3	4,205
	(0.013)				
Road transport, baseline	0.139***	[0.07, 0.21]	[0.08, 0.26]	10.7	4,213
	(0.035)				

Notes: Each row is a two-stage least squares regression of cumulative extra CO_2 over horizons 0 to 8 (5 where stated), in kg, on cumulative extra fossil rent revenue over the same horizons, in US dollars, both relative to $t-1$ and divided by GDP in $t-1$, with the windfall shock as the instrument and two lags of the shock and lagged changes in emissions and rents as controls; country and year fixed effects; the top and bottom one per cent of each cumulative variable are trimmed. The sample is country-years with total fossil CO_2 of at least 1 Mt in $t-1$, shocks from 1981 to 2024; the fuel-pricing rows split it by whether the gasoline net tax (Ross et al. 2017, extended) averaged below zero over $t-5$ to $t-1$. Standard errors in parentheses are two-way clustered by country and year; stars are from Anderson-Rubin tests of a zero carbon content, and the Anderson-Rubin set collects every value not rejected at the five per cent level, which remains valid when the first stage is weak. The fuel rows instrument revenue with one fuel's exposure times its price surprise at a time; with all three as instruments the estimate is 0.549 and the Sargan test of equal carbon content across fuels gives p = 0.078 (not robust to clustering). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A6: EIA energy CO_2 and measured fuel and electricity consumption

Horizon	Energy CO_2 (EIA)	Oil consumption	Gasoline and diesel	Gas consumption	Electricity consumption	Fossil power generation
Panel A. Windfall shock (price innovations)						
Pre-period, $h = -2$	0.002	0.040	0.048	0.060	0.051	−0.027
	(0.053)	(0.063)	(0.063)	(0.080)	(0.039)	(0.086)
$h = 0$	0.104	0.131**	0.226***	0.039	−0.008	−0.045
	(0.081)	(0.062)	(0.079)	(0.145)	(0.039)	(0.108)
$h = 2$	0.326**	0.435***	0.564***	0.068	0.211*	0.206
	(0.129)	(0.101)	(0.147)	(0.204)	(0.123)	(0.193)
$h = 3$	0.346**	0.522***	0.625***	0.104	0.235	0.427
	(0.163)	(0.160)	(0.210)	(0.217)	(0.149)	(0.265)
$h = 5$	0.476***	0.604**	0.755**	0.604**	0.395**	0.500
	(0.171)	(0.225)	(0.325)	(0.255)	(0.190)	(0.330)
$h = 8$	0.456*	0.572*	0.882**	0.255	0.535**	0.484
	(0.234)	(0.288)	(0.325)	(0.366)	(0.223)	(0.342)
Observations ($h = 5$)	4,745	4,745	4,745	3,111	4,745	4,694
Panel B. Windfall shock instrumented by oil supply news, outside OPEC						
Pre-period, $h = -2$	0.240	0.098	0.159*	0.084	−0.053	−0.323
	(0.205)	(0.092)	(0.086)	(0.171)	(0.090)	(0.252)
$h = 0$	0.401**	0.501***	0.553***	0.398*	0.233	0.370
	(0.187)	(0.122)	(0.132)	(0.228)	(0.153)	(0.335)
$h = 2$	0.747***	0.479**	0.453**	1.139**	0.442*	0.633
	(0.190)	(0.227)	(0.216)	(0.531)	(0.238)	(0.472)
$h = 3$	0.411***	0.204	0.236	1.147	0.396	0.498
	(0.070)	(0.269)	(0.309)	(0.702)	(0.268)	(0.531)
$h = 5$	0.200	−0.042	0.437	1.041**	0.662*	0.691
	(0.321)	(0.282)	(0.307)	(0.433)	(0.358)	(0.623)
$h = 8$	−0.034	−0.180	0.259	0.072	0.155	0.210
	(0.468)	(0.537)	(0.587)	(0.187)	(0.351)	(0.420)
First-stage F ($h = 5$)	18.8	18.9	18.9	21.0	19.1	18.9

Notes: Outcomes are logs of series from the US Energy Information Administration international energy data (bulk file, October 2026): energy-related CO_2 computed by the EIA, total consumption of petroleum and other liquids, the sum of motor gasoline and distillate fuel oil consumption, dry natural gas consumption, net electricity consumption and net electricity generation from fossil fuels. Panel A reports local projections on the windfall shock as in Tables 2 and 3; Panel B instruments the windfall shock with oil rents times time-aggregated Känzig (2021) oil supply news in countries outside OPEC. The sample is country-years with total fossil CO_2 (EDGAR) of at least 1 Mt in $t-1$, shocks from 1981 to 2024. Country and year fixed effects, two lags of the shock and of the outcome change; standard errors in parentheses are two-way clustered by country and year. EIA energy CO_2 and EDGAR fossil CO_2 net of process and fugitive emissions correlate at 0.989 in log levels and 0.392 in annual log changes. The carbon content of windfall revenue measured with EIA energy CO_2, as in Table 4, is 0.510 kg per dollar (standard error 0.159, first-stage F 11.2, Anderson-Rubin p = 0.004). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A7: Supply-news responses with four lags and with the simple annual sum of news

Horizon	Total CO_2	Public investment	Household consumption	Cement CO_2	Buildings CO_2	Road transport CO_2
Panel A. Windfall shock instrumented by time-aggregated supply news, four lags						
Pre-period, $h = -3$	-0.172	-1.064^{*}	-0.392^{**}	-0.492^{*}	-1.220	-0.031
	(0.108)	(0.542)	(0.186)	(0.286)	(0.936)	(0.164)
Pre-period, $h = -2$	-0.172	-0.654^{**}	-0.182^{**}	-0.523^{***}	-0.586	-0.174
	(0.119)	(0.267)	(0.086)	(0.164)	(0.487)	(0.119)
$h = 0$	0.224	1.155	0.259^{**}	0.405	0.337	0.524^{*}
	(0.179)	(0.723)	(0.111)	(0.319)	(0.360)	(0.298)
$h = 2$	0.276^{*}	2.385^{**}	0.432^{**}	1.051^{*}	0.871	0.568^{**}
	(0.146)	(1.023)	(0.211)	(0.529)	(0.573)	(0.240)
$h = 3$	0.281^{**}	1.897^{***}	0.545^{**}	1.117	0.986^{*}	0.476^{**}
	(0.104)	(0.554)	(0.255)	(0.753)	(0.545)	(0.220)
$h = 5$	0.015	0.221	0.425^{*}	1.989^{*}	1.810^{**}	0.505^{*}
	(0.093)	(0.943)	(0.223)	(1.145)	(0.727)	(0.290)
First-stage F ($h = 3$)	28.3	29.6	13.1	20.5	25.1	28.3
Panel B. Reduced form, oil rents times the simple annual sum of supply news						
Pre-period, $h = -3$	-0.031^{**}	-0.092	-0.024^{*}	0.004	-0.154^{**}	-0.018
	(0.014)	(0.083)	(0.013)	(0.050)	(0.069)	(0.020)
Pre-period, $h = -2$	-0.025^{***}	-0.054	-0.005	-0.063	-0.089	-0.016
	(0.004)	(0.049)	(0.013)	(0.039)	(0.057)	(0.014)
$h = 0$	0.010	0.115^{**}	0.028^{**}	0.044^{**}	-0.004	0.025
	(0.011)	(0.055)	(0.012)	(0.016)	(0.029)	(0.021)
$h = 2$	0.026	0.249^{**}	0.058^{**}	0.146^{**}	0.106^{*}	0.072^{**}
	(0.024)	(0.114)	(0.023)	(0.068)	(0.054)	(0.034)
$h = 3$	0.016	0.265^{**}	0.064^{**}	0.196^{***}	0.126^{*}	0.045
	(0.024)	(0.122)	(0.028)	(0.068)	(0.066)	(0.039)
$h = 5$	0.002	0.070	0.069^{**}	0.225^{**}	0.169^{**}	0.054
	(0.016)	(0.114)	(0.028)	(0.111)	(0.067)	(0.038)
Observations ($h = 5$)	4,186	3,214	3,285	3,840	4,148	4,186

Notes: Panel A repeats the instrumental-variables local projections of Panel B of Tables 2 and 3 with four lags of the windfall shock and of the outcome change. Panel B reports reduced forms in which the shock is oil rents times the simple annual sum of the monthly Känzig (2021) news shocks, the timing used before time aggregation; its coefficients are per unit of that variable. Both panels use countries outside OPEC. With the simple annual sum as the instrument instead, the first-stage F statistic at $h = 3$ lies between 0.5 and 2.5 for these outcomes. The sample is country-years with total fossil CO_2 of at least 1 Mt in $t - 1$, shocks from 1981 to 2024. Country and year fixed effects; standard errors in parentheses are two-way clustered by country and year. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A8: Supply-news responses averaged over horizons

Specification	Total CO_2 (EDGAR)	Energy CO_2 (EIA)	Road transport CO_2	Cement CO_2	Power CO_2
Supply news as instrument, years 0 to 3	0.270^{*}	0.550^{***}	0.745^{***}	1.167^{**}	0.264
	(0.151)	(0.107)	(0.241)	(0.509)	(0.276)
Supply news as instrument, years 0 to 5	0.172	0.484^{***}	0.735^{***}	1.578^{**}	0.161
	(0.152)	(0.115)	(0.255)	(0.722)	(0.287)
Supply news as instrument, four lags, years 0 to 5	0.260^{**}	0.476^{***}	0.602^{**}	1.259^{*}	-0.083
	(0.100)	(0.089)	(0.233)	(0.720)	(0.190)
Price surprises (OLS), same countries, years 0 to 5	0.382^{***}	0.444^{**}	0.505^{***}	0.706^{*}	-0.103
	(0.092)	(0.188)	(0.158)	(0.378)	(0.178)
First-stage F (years 0 to 5)	18.8	18.8	19.0	16.3	18.9

Notes: The outcome is the average over horizons 0 to H of the cumulative log change from $t-1$ to $t+h$. Rows 1 to 3 instrument the windfall shock with oil rents times time-aggregated Känzig (2021) supply news; the last row is OLS on the same countries outside OPEC. Controls are two (or four) lags of the windfall shock and of the outcome change, with country and year fixed effects; standard errors in parentheses are two-way clustered by country and year. Over years 0 to 5, the p-values of the supply-news estimates are 0.266 for total CO_2 (EDGAR), 0.014 for total CO_2 with four lags and below 0.001 for EIA energy CO_2. EIA energy CO_2 is computed by the US Energy Information Administration from measured fuel consumption. The sample is country-years with total fossil CO_2 of at least 1 Mt in $t-1$, shocks from 1981 to 2024. * $p<0.10$, ** $p<0.05$, *** $p<0.01$.

Table A9: Identification checks for road transport, buildings and household consumption

Specification	Road CO_2, $h = 3$	Road CO_2, $h = 5$	Buildings CO_2, $h = 3$	Buildings CO_2, $h = 5$	Consumption, $h = 3$	Consumption, $h = 5$
Baseline	0.509^{***}	0.760^{***}	0.666^{***}	0.916^{***}	0.376^{***}	0.445^{***}
	(0.177)	(0.223)	(0.227)	(0.302)	(0.080)	(0.128)
Global-cycle controls	0.495^{***}	0.867^{***}	0.658^{***}	0.908^{***}	0.306^{***}	0.355^{**}
	(0.117)	(0.143)	(0.229)	(0.211)	(0.098)	(0.165)
Region × year FE and global-cycle controls	0.581^{***}	0.967^{***}	0.810^{***}	0.923^{***}	0.267^{**}	0.308^{**}
	(0.147)	(0.183)	(0.266)	(0.241)	(0.102)	(0.147)
Excluding OPEC and Russia	0.486^{***}	0.470^{**}	1.121^{***}	1.532^{**}	0.335^{*}	0.366^{*}
	(0.165)	(0.231)	(0.335)	(0.568)	(0.180)	(0.202)
Excluding sanctioned economies and Kuwait	0.720^{***}	0.822^{***}	0.895^{***}	1.194^{***}	0.359^{***}	0.503^{***}
	(0.187)	(0.300)	(0.264)	(0.363)	(0.095)	(0.180)
Exposure measured $t-15$ to $t-11$	0.840^{***}	1.117^{***}	0.898^{**}	1.314^{**}	0.334^{***}	0.440^{***}
	(0.247)	(0.361)	(0.347)	(0.524)	(0.088)	(0.132)
Placebo: metal rents × oil innovation	−0.732	−1.457	1.192	1.428	0.048	0.252
	(0.981)	(1.300)	(1.191)	(1.624)	(0.322)	(0.492)

Notes: Each cell is the coefficient on the windfall shock from a separate local projection of the cumulative log change in road transport CO_2, buildings CO_2 or real household consumption; the specifications are those of Table 6. The sample is country-years with total fossil CO_2 of at least 1 Mt in $t-1$, shocks from 1981 to 2024. Estimation is by OLS local projections with country and year fixed effects and two lags of the shock and of the outcome change; standard errors in parentheses are two-way clustered by country and year. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A10: Randomisation inference ($h = 5$)

Outcome	Estimate	p, innovation years permuted	Draws	p, exposure permuted within regions	Draws
Total fossil CO_2	0.265	0.060	500	0.010	300
Road transport CO_2	0.760	0.018	500	0.000	300
Cement process CO_2	0.906	0.004	500	0.010	300

Notes: The shock-level test permutes the vector of oil, gas and coal price innovations across years, keeping each country's exposure fixed, and re-estimates the baseline local projection; the exposure-level test permutes the exposure histories (oil, gas and coal rents) across countries within World Bank regions. The p-values are the shares of placebo estimates at least as large in absolute value as the estimate. The sample is country-years with total fossil CO_2 of at least 1 Mt in $t-1$, shocks from 1981 to 2024. No significance stars are used.

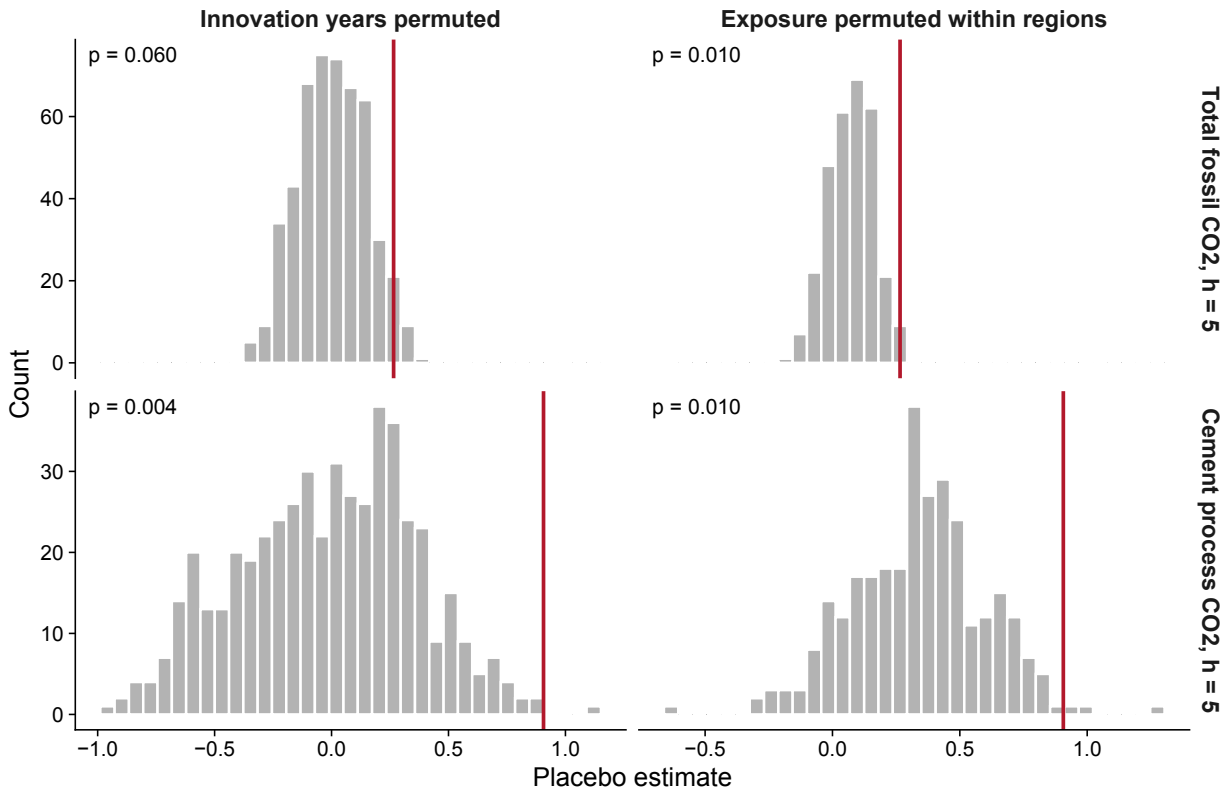


Figure A1: Randomisation inference for the five-year responses. Vertical lines: estimates with the true assignment; p-values are two-sided

Dropping, in turn, each year of price surprises leaves the five-year total CO_2 response significant at five per cent in every case, and dropping each major price episode leaves it between 0.17 and 0.32 and significant except when the 2014 to 2016 price collapse or all years after 2014 are left out (Table A11).

Table A11: Leaving out years and price episodes

	Total CO_2, $h = 3$	Total CO_2, $h = 5$	Cement CO_2, $h = 5$	Carbon content
Panel A. Leave one shock year out				
Range of estimates	0.172 to 0.260	0.208 to 0.307	0.816 to 1.060	0.416 to 0.579
Significant at 5 per cent	34 of 37	37 of 37	37 of 37	36 of 37
Significant at 10 per cent	37 of 37	37 of 37	37 of 37	36 of 37
Weakest case (year dropped, p)	2015, 0.079	2015, 0.039	2015, 0.002	1986, 0.100
Panel B. Episodes dropped				
1985 to 1987 oil price collapse	0.275***	0.271**	0.871***	0.456*
	(0.100)	(0.101)	(0.229)	(0.188)
1990 to 1991 Gulf war	0.244**	0.276***	0.885***	0.522**
	(0.096)	(0.074)	(0.161)	(0.141)
1998 to 1999 price trough	0.244**	0.273***	0.920***	0.576***
	(0.105)	(0.072)	(0.154)	(0.162)
2003 to 2008 boom	0.206*	0.171***	0.613***	0.610**
	(0.107)	(0.054)	(0.205)	(0.206)
2009 crash	0.241*	0.285**	1.060***	0.506**
	(0.120)	(0.112)	(0.175)	(0.192)
2014 to 2016 price collapse	0.135	0.202	0.838**	0.521*
	(0.100)	(0.129)	(0.338)	(0.257)
2020 to 2022 pandemic and energy crisis	0.255**	0.265***	0.906***	0.522**
	(0.100)	(0.073)	(0.156)	(0.178)
Whole 2003 to 2014 boom	0.216	0.319***	0.677***	0.699***
	(0.138)	(0.082)	(0.214)	(0.261)
All years before 2000	0.324**	0.273**	0.988**	0.489**
	(0.118)	(0.124)	(0.386)	(0.150)
All years after 2014	0.213*	0.225*	0.897***	0.525*
	(0.115)	(0.121)	(0.301)	(0.251)

Notes: Panel A re-estimates each specification dropping, in turn, every country-year whose shock year equals a given year, after building leads, lags and lagged shocks on the full panel, so that the price surprise of that year no longer contributes to identification. Panel B drops all shock years in the listed episode. Columns 1 to 3 are local projections of cumulative log CO_2 as in Tables 3 and 5; the last column is the carbon content of windfall revenue (kg CO_2 per US dollar) as in Table 4, with stars from Anderson-Rubin tests. In Panel B, the p-value of the five-year total CO_2 response is 0.126 when the 2014 to 2016 price collapse is dropped and 0.072 when all years after 2014 are dropped. The sample is country-years with total fossil CO_2 of at least 1 Mt in $t-1$, shocks from 1981 to 2024. Country and year fixed effects; standard errors in parentheses are two-way clustered by country and year. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Adding the sum of exposure as a control, the adjustment recommended for exposure shares that do not sum to one when the shocks are as good as randomly assigned (Borusyak

et al. 2022), leaves the cement response significant (1.06 after five years) and the total CO_2 response significant at the ten per cent level, at 0.19 (Table A12). Forecast errors make that assumption plausible but do not guarantee it. The same table computes standard errors from the equivalent regression on fuel-year price surprises and reports the more demanding specification with exposure-quintile by year fixed effects. That specification leaves only the variation that comes from differences between oil, gas and coal surprises within a year; there the total CO_2 estimate falls to 0.17 and loses significance, and the cement estimate falls.

Table A12: Incomplete exposure shares and shock-level inference

	Total CO_2, $h = 3$	Total CO_2, $h = 5$	Cement CO_2, $h = 5$
Panel A. Exposure controls			
Sum of exposure S as a control	0.187*	0.191*	1.060***
	(0.102)	(0.099)	(0.195)
Exposure quintile × year fixed effects	0.023	0.172	0.323
	(0.149)	(0.145)	(0.450)
Panel B. Shock-level inference (sum of exposure controlled)			
Coefficient	0.187**	0.191**	1.060***
Exposure-robust standard error	(0.083)	(0.077)	(0.262)
Conventional standard error, country-year regression	(0.102)	(0.099)	(0.195)
Fuel-year shocks	111	105	105
Years	39	37	37
First-stage F, shock level	173.6	321.1	275.7

Notes: Fossil rents shares do not sum to one, so the windfall shock also varies with the level of exposure. Panel A adds the sum of exposure S as a control, the adjustment recommended by Borusyak et al. (2022) when shocks are as good as randomly assigned (the innovations are recursive forecast errors, so a zero conditional mean and independence from other determinants of the outcomes are identifying assumptions), and, as a more demanding check, fixed effects for quintiles of S interacted with years, which leave only the variation that comes from differences between oil, gas and coal price surprises in a given year. Panel B reports the equivalent fuel-year regression of exposure-weighted residualised outcomes on the residualised windfall shock, instrumented by the price innovation and weighted by total exposure, with standard errors clustered by year (Borusyak et al. 2022; Adão et al. 2019); the coefficient equals that of the country-year regression with S controlled. The p-values of the shock-level coefficients in Panel B are 0.029 for total CO_2 at $h = 3$, 0.018 for total CO_2 at $h = 5$ and below 0.001 for cement CO_2 at $h = 5$. The sample is country-years with total fossil CO_2 of at least 1 Mt in $t - 1$, shocks from 1981 to 2024. Country and year fixed effects in the country-year regressions, with standard errors two-way clustered by country and year. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Matching each country's oil, gas and coal exposure to the price benchmark of the market it sells into adds variation within years (Table A13). The benchmarks are WTI crude, US natural gas and South African coal in the Americas; Dubai crude and Japanese liquefied natural gas in Asia and the Middle East, with Australian coal in South Asia and East Asia and South African coal in the Middle East; and Brent crude, European gas and South

African coal elsewhere. Regional surprises are highly correlated with world surprises, so the estimates change little, but with exposure quintile by year fixed effects the road transport response becomes significant at the five per cent level, while total CO_2 ($p = 0.12$) and cement remain insignificant.

Table A13: Windfall responses with regional price benchmarks ($h = 5$)

Specification	Total CO_2	Cement CO_2	Road transport CO_2	Buildings CO_2
World prices (main text)	0.265***	0.906***	0.760***	0.916***
	(0.073)	(0.156)	(0.223)	(0.302)
Regional benchmarks	0.262***	0.886***	0.732***	0.847***
	(0.077)	(0.147)	(0.217)	(0.287)
Regional benchmarks, sum of exposure controlled	0.189*	1.034***	0.681**	0.895***
	(0.100)	(0.189)	(0.258)	(0.295)
Regional benchmarks, exposure quintile × year FE	0.196	0.373	0.626**	0.279
	(0.123)	(0.451)	(0.236)	(0.377)
Regional benchmarks, region × year FE	0.250**	0.775***	0.825***	0.907***
	(0.095)	(0.211)	(0.257)	(0.289)

Notes: Each cell is the five-year cumulative log response to a unit windfall shock. In the regional rows, each country's oil, gas and coal exposure is matched to the price benchmark of the market it sells into: WTI, US natural gas and South African coal in the Americas; Dubai, Japanese LNG and Australian coal in the Middle East outside North Africa, South Asia and East Asia and the Pacific (Australian coal only in the last two); Brent, European natural gas and South African coal elsewhere. Innovations are real-time AR(2) forecast errors for each benchmark (World Bank Pink Sheet, deflated by the US CPI), replaced by the world innovation where a benchmark series is too short. The exposure quintile × year row absorbs everything common to countries with similar exposure in a year, so identification comes from differences between fuels and between regional markets. In the exposure quintile × year row, the p-values are 0.121 for total CO_2, 0.414 for cement CO_2, 0.012 for road transport CO_2 and 0.465 for buildings CO_2. Country and year fixed effects unless stated, two lags of the shock and of the outcome change; standard errors in parentheses are two-way clustered by country and year. The sample is country-years with total fossil CO_2 of at least 1 Mt in $t - 1$, shocks from 1981 to 2024. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

F The construction channel

EDGAR cement emissions agree with the Global Carbon Project series in log levels (correlation 0.98) and less closely in annual log changes (0.66), with the weakest agreement in changes for country-years based on official statistics or clinker data (Table A14).

Windfalls do not expand the building footprint measured by the Global Human Settlement Layer (Pesaresi et al. 2024): none of the estimates for total built-up surface is significant

Table A14: Validation of cement emissions: EDGAR against the Global Carbon Project dataset

GCP source flag	Country-years	Correlation, log levels	Correlation, annual log changes
All country-years	4,995	0.976	0.657
CRT	2,004	0.992	0.894
GNR_CR	1,494	0.987	0.601
Interpolated	886	0.939	0.601
Official	296	0.883	0.123
Clinker	209	0.986	0.238
Country specific	79	0.997	0.608
Apparent clinker	24	0.979	0.662

Notes: The table compares EDGAR 2026 cement process CO_2 with the Global Carbon Project cement dataset (release 260526) by the source flag of each country-year in the analysis sample: CRT are UNFCCC common reporting tables, GNR_CR industry data from the GCCA Getting the Numbers Right database, Official national statistics, Clinker estimates from clinker production, Country specific and Apparent clinker other country estimation methods, and Interpolated fills gaps between reported years; correlations are Pearson coefficients. The Global Carbon Project series in Table 5 excludes interpolated years. No significance tests are reported.

(Table A15). I sum the 1 km grids of total and non-residential built-up surface (Pesaresi 2023) within Natural Earth country borders (Natural Earth 2026) for every five-year epoch from 1975 to 2020. Because only the 1975, 1990, 2000, 2014 and 2018 layers rest on satellite observations, the table reports both the five-year epochs and the intervals between observed layers. The non-residential footprint falls over observed intervals in one specification and does not respond in the five-year design.

Before turning to windfalls I tested whether short-run changes in the price of fuel or energy move carbon-intensive production. None of these tests finds that existing heavy industry responds to short-run prices, which is the putty-clay assumption of Section 2.3 (Table A16).

The plant-level evidence uses a monthly panel of activity at 4,715 cement and steel plants built from Sentinel-5P TROPOMI tropospheric nitrogen dioxide columns (Copernicus Sentinel-5P 2021; van Geffen et al. 2022), processed in Google Earth Engine (Gorelick et al. 2017), comparing the column within 5 km of each plant with a ring 20 to 40 km away (Beirle et al. 2019; Veefkind et al. 2012). Plant locations come from the Spatial Finance Initiative asset databases for cement and for iron and steel (McCarten et al. 2021; Tkachenko et al. 2023). The measure tracks official cement output at annual frequency and shows no gap at placebo sites without plants (Table A17). It does not detect mandated winter kiln shutdowns in northern China, and the update of the retrieval processor to version 2.4 in July 2022 raised the signal at bright sites.

Table A15: Built-up surface after windfalls (Global Human Settlement Layer)

	Total, five-year epochs	Total, five-year epochs	Non-residential, five-year epochs	Total, observed intervals	Total, observed intervals	Non-residential, observed intervals
Windfall	0.006	−0.005	−0.020	−0.191	−0.019	−0.456**
	(0.024)	(0.006)	(0.022)	(0.145)	(0.017)	(0.179)
Shock	Innovation	Supply news	Innovation	Innovation	Supply news	Innovation
Countries	151	135	151	145	129	145
Observations	866	761	866	393	347	393

Notes: The outcome is the log change in national built-up surface, the building footprint excluding roads and other open infrastructure, from the Global Human Settlement Layer GHS-BUILT-S R2023A 1 km grids summed within country borders (Pesaresi et al. 2024). Columns 1 to 3 use five-year epochs from 1975 to 2020 and regress the five-year change on the sum of windfall shocks over the same five years, its lag and the lagged change, with country and epoch fixed effects. Only the 1975, 1990, 2000, 2014 and 2018 layers rest on satellite observations, and the remaining epochs are interpolated by the producer, so the last three columns use the intervals between observed layers (1975 to 1990, 1990 to 2000, 2000 to 2015 and 2015 to 2020, the latter two approximating the 2014 and 2018 observations) and regress the annualised log change on the mean annual shock over the interval, with country and interval fixed effects; supply-news columns use countries outside OPEC. The sample is country-years with total fossil CO_2 of at least 1 Mt in $t-1$, shocks from 1981 to 2024. Standard errors in parentheses are clustered by country. * $p<0.10$, ** $p<0.05$, *** $p<0.01$.

Table A16: Short-run fuel and energy price designs

Design	Outcome	Result
Administered fuel price hikes (228 events, 1995 to 2023)	Carbon composition of manufacturing (TiVA, exports)	No effect up to eight years; four estimators
2022 energy shock, fixed vs market retail prices	Plant NO_2 at 2,901 plants	Gap of 0.018 (p = 0.010), but no dose in world prices, retail prices or IMF subsidies
Countries that held prices vs passed the shock through	Plant NO_2	Similar gaps (0.016 and 0.021)
Kilns vs grinding plants, country × month FE	Plant NO_2	0.002 (p = 0.50); measure fails a positive control
EU energy price shock by sector energy intensity	EU imports from non-EU producers	Value up, tonnes unchanged; diversion from sanctioned suppliers

Notes: The table summarises tests, documented in the replication package, that ask whether changes in the price of fuel or energy move carbon-intensive production in the short run. Estimates come from country-sector panels built from OECD TiVA (OECD 2023) and CEPII BACI (Gaulier and Zignago 2010), satellite plant panels and EU trade flows (Eurostat 2026), with the fixed effects documented in the package. Together they motivate the putty-clay assumption of Section 2.3: installed carbon-intensive capacity does not respond to short-run price changes, and the table reports no new estimates.

Table A17: Validation of the satellite measure of plant activity

Check	Result
Validation against official cement output (U.S. Geological Survey 2026; 14 producers, 2019 to 2025)	Log output on capacity-weighted plant signal, country and year FE: 2.729 (p = 0.007)
Monthly validation (Eurostat 2026; Federal Reserve Bank of St. Louis 2026; 7 countries)	Weak: 12-month changes 0.543 (p = 0.104) for cement
TROPOMI processor v2.4 (17 July 2022) and surface albedo (MODIS MCD43A3; Schaaf and Wang 2021)	Signal rises with site albedo after the switch: 0.012 per standard deviation (p = 0.007); controlling for it removes the plant-level windfall effect (Table A18)
Placebo sites with no plant within 40 km	Gap at placebo sites 0.000 (p = 0.92); plant increment over own placebo 0.018 (p = 0.016)
Positive control: winter kiln shutdowns in northern China	Not detected: 0.001 (standard error 0.023)

Notes: The plant signal is the log ratio of tropospheric NO_2 within 5 km of a plant to a 20 to 40 km ring (Sentinel-5P TROPOMI). Validation regressions use country and year fixed effects with standard errors clustered by country; the placebo test uses 1,826 sites 50 to 100 km from plants in the same countries. The signal tracks annual cement output and passes the placebo test, but it cannot resolve month-level on and off states of individual kilns, so plant results are read at annual or multi-month horizons. No significance stars are used.

Plants in countries that held retail fuel prices fixed through 2017 to 2021 are mostly in North Africa, the Middle East and Indonesia, with smaller groups in Latin America and sub-Saharan Africa (Figure A2). Because the retrieval change coincides with the 2022 energy price shock, the post-2022 gap in activity between plants in countries with fixed retail fuel prices and plants in countries with market prices cannot be separated from the artefact (Figures A2 and A3). The apparent effect of windfalls on plant activity also disappears once the artefact is controlled for, so the plant data neither confirm nor contradict the country-level results (Table A18).

G Anticipation and disasters

Giant oil and gas discoveries are news about future rents (Arezki et al. 2017), yet they raise neither cement nor total emissions (Table A19). The public investment share reported by Arezki et al. (2017) rises four to eight years after a discovery, when production typically begins, while public investment in the IMF dataset does not respond, so the two series disagree.

Large climate disasters recorded in EM-DAT (Delforge et al. 2025) raise cement emissions in economies with high fossil rents, which would make disasters windows for investment, but

Table A18: Plant activity after windfalls and the 2022 change in the satellite retrieval

Specification	Windfall coefficient	Standard error	Plant-months
Baseline (plant and month FE)	0.153**	(0.064)	379,551
+ v2.4 era × site albedo and albedo contrast	0.016	(0.061)	379,551
+ albedo controls, plant × calendar-month FE	0.016	(0.048)	379,531
Months before the v2.4 switch only (2019 to June 2022)	−0.089	(0.060)	147,815
Excluding the brightest quarter of sites	0.029	(0.098)	284,644
Plant minus its own placebo site (2019 to 2023)	0.196**	(0.078)	88,505
Placebo sites only (2019 to 2023)	−0.024	(0.047)	88,505

Notes: The dependent variable is the log ratio of tropospheric NO_2 within 5 km of a cement or steel plant to a 20 to 40 km ring (Sentinel-5P TROPOMI, monthly, 2019 to 2026 without March to December 2020); the regressor is each country's 2015 to 2019 fossil rents share times the trailing 12-month log real world fuel price. Estimation is by OLS with plant and month fixed effects (plant by calendar month where stated) and the COVID-19 stringency index of the Oxford COVID-19 Government Response Tracker (Hale et al. 2021); standard errors are clustered by country. Processor version 2.4.0 of the TROPOMI NO_2 retrieval (operational from 17 July 2022) changed the surface reflectivity inputs, which raises the plant signal at bright sites; once the post-switch period is interacted with site albedo, the windfall coefficient is close to zero, so the plant data do not independently confirm the country-level results. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

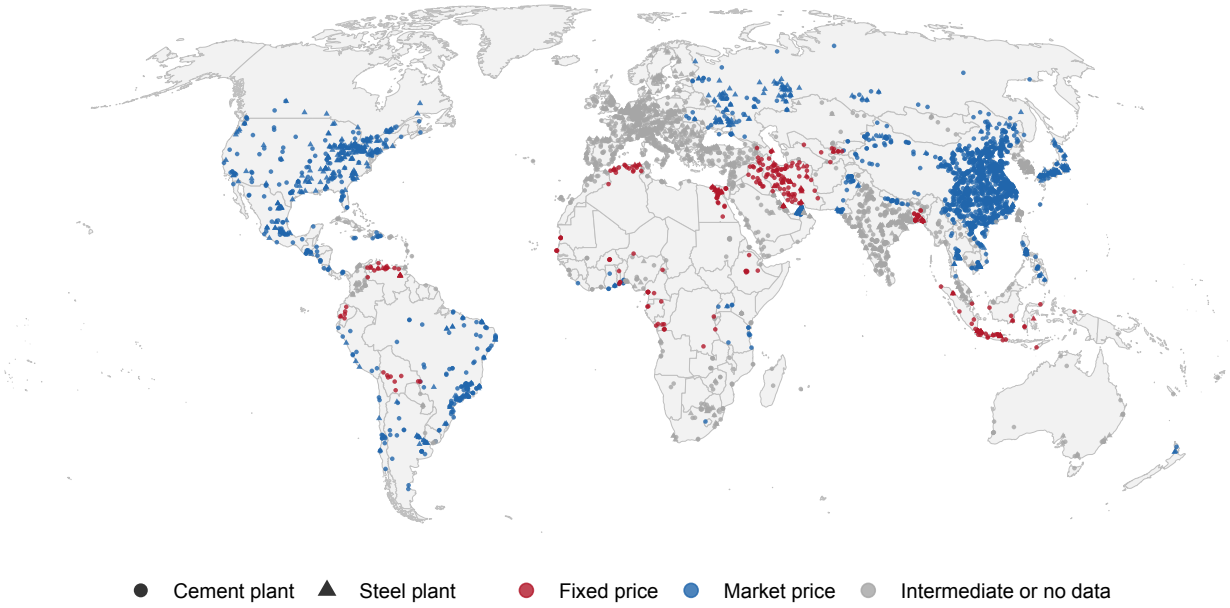


Figure A2: Cement and steel plants by retail fuel price regime

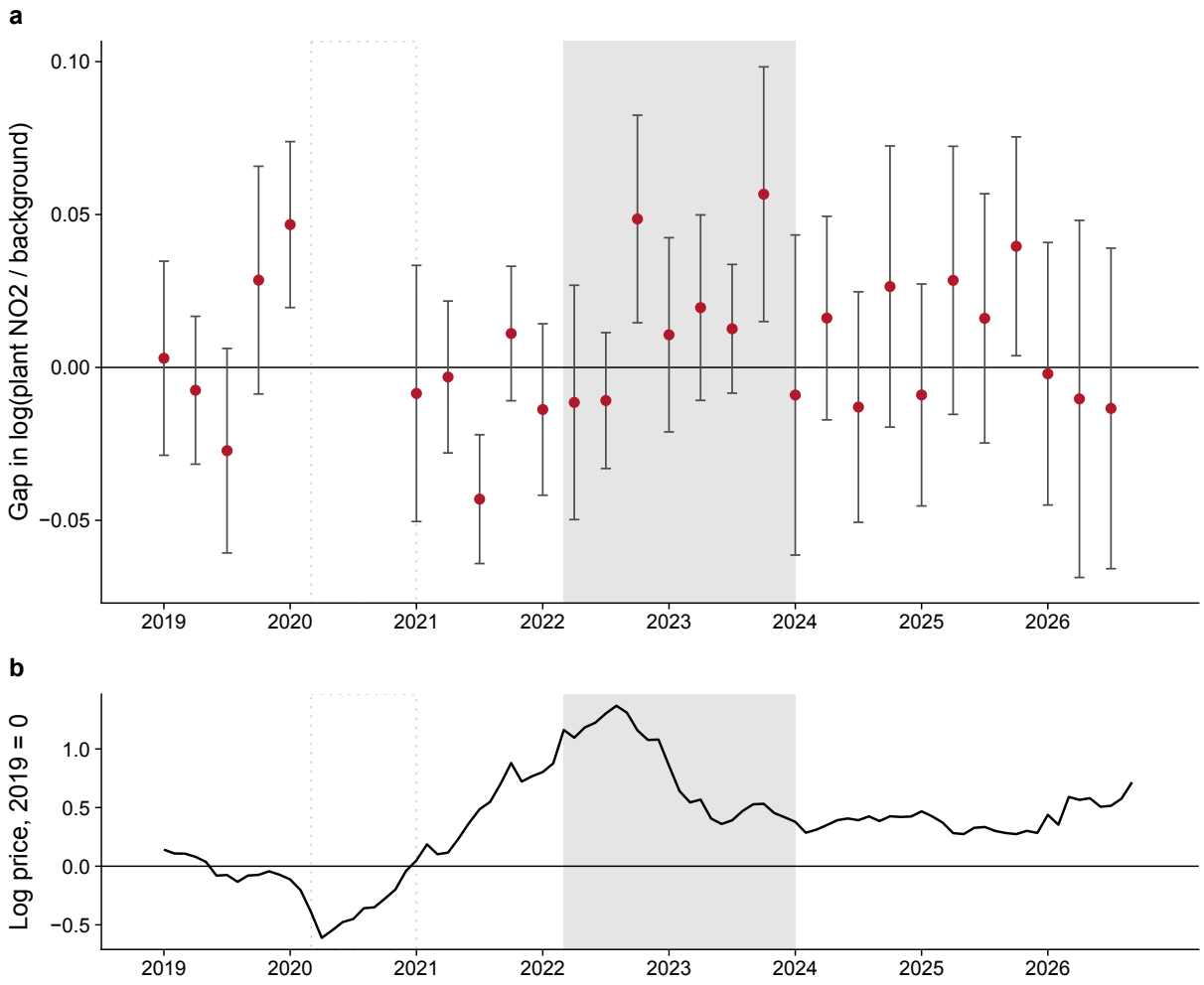


Figure A3: Gap in activity between fixed-price and market-price plants (a) and world energy prices (b) around the 2022 energy price shock. Whiskers: 95 per cent confidence intervals clustered by country; dotted box: 2020, excluded from estimation

Table A19: Giant oil and gas discoveries

Horizon	Cement CO_2 (EDGAR)	Cement output	Total fossil CO_2	Public inv. share (ARS)	Public investment (IMF)	Real GDP
Pre-period, $h = -2$	0.061	−0.021	0.021	−0.065**	−0.033	−0.018
	(0.061)	(0.033)	(0.028)	(0.025)	(0.066)	(0.019)
$h = 0$	−0.237	−0.063	0.009	−0.012	−0.026	−0.004
	(0.147)	(0.048)	(0.014)	(0.047)	(0.067)	(0.012)
$h = 4$	0.040	0.010	−0.004	0.625***	−0.065	0.008
	(0.122)	(0.080)	(0.033)	(0.200)	(0.120)	(0.029)
$h = 6$	0.100	0.147*	0.036	0.974***	−0.151	0.016
	(0.072)	(0.083)	(0.038)	(0.284)	(0.125)	(0.033)
$h = 8$	0.131	0.165*	0.023	0.379*	−0.162	0.030
	(0.086)	(0.082)	(0.036)	(0.188)	(0.120)	(0.033)
Observations ($h = 6$)	3,339	3,235	3,644	798	3,367	3,610

Notes: The shock is the log of one plus the net present value of giant oil and gas discoveries as a share of GDP (Arezki et al. 2017), 1970 to 2012; production from a giant field typically starts four to six years after discovery. The sample is country-years with total fossil CO_2 of at least 1 Mt in $t - 1$, shocks from 1981 to 2024. Estimation is by OLS local projections with country and year fixed effects and two lags of the shock and of the outcome change; standard errors in parentheses are two-way clustered by country and year. The ARS public investment share is the series of Arezki, Ramey and Sheng (2017), from the IMF World Economic Outlook; the IMF column is public investment from the Investment and Capital Stock Dataset. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

only three such disasters occurred (Table A20).

Table A20: Climate disasters, fossil rents and cement emissions

	$h = 2$	$h = 3$	$h = 5$
Average effect, people affected (per 10% of population)			
Cement CO_2 (GCP)	0.026	−0.018	0.009
	(0.023)	(0.053)	(0.032)
Public investment	0.000	−0.006	−0.008
	(0.026)	(0.035)	(0.039)
Large disaster interacted with fossil rents share S			
Cement CO_2: large disaster	−0.037	−0.246	−0.137
	(0.110)	(0.246)	(0.121)
Cement CO_2: large disaster × S	6.163***	10.822**	9.128***
	(2.029)	(3.916)	(2.375)
Public investment: large disaster × S	−1.143	−1.197	−5.743
	(1.741)	(3.169)	(6.266)
Large disasters in countries with S at least 0.10	3	3	3

Notes: The shocks are climate-related disasters (climatological, hydrological, meteorological) recorded in EM-DAT, 2000 to 2022: people affected per 10 per cent of the population, and a large-event indicator for the top 5 per cent of country-years; the interaction model adds the lagged fossil rents share S, its interaction with the disaster indicator and its lags, and the windfall shock. The sample is country-years with total fossil CO_2 of at least 1 Mt in $t-1$, shocks from 1981 to 2024. Estimation is by OLS local projections with country and year fixed effects and two lags of the shock and of the outcome change; standard errors in parentheses are two-way clustered by country and year. Few large disasters hit economies with rents of at least 10 per cent of GDP, so the interaction rests on few events. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.